

Parity equidistribution of nested floor powers, with descent applications to the Juggler map

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Abstract

For odd n set $m = \lfloor n^{3/2} \rfloor$, $v = \lfloor m^{3/2} \rfloor$, and so on: the integer chains obtained by iterating $x \mapsto \lfloor x^{3/2} \rfloor$ and $x \mapsto \lfloor \sqrt{x} \rfloor$ in a prescribed pattern. We prove parity equidistribution, with power savings, for every such pattern of depth at most four over odd starts. The obstacle at depth two and beyond is that the naive expansion of $m^{3/2}$ leaves the sawtooth $\{n^{3/4}\}$ with an amplitude that grows like $n^{3/4}$, which defeats the classical van der Corput method. The proof reorganizes the phase by an exact linearization — $m^{3/2} = \frac{3}{2}mn^{3/4} - \frac{1}{2}n^{9/4} + O(n^{-3/4})$ with a one-signed remainder, a Taylor rewrite that keeps the integer m linear with a smooth coefficient — and funnels the deeper patterns into a single *kernel*: the exponential sum of the level-2 floor defect $\{\lfloor n^{3/2} \rfloor^{3/2}\}$ against smooth weights of scale $n^{9/8}$. The central result (Theorem 5.3) is a power-saving bound for that kernel, $K_c \ll P^{1-1/96+\varepsilon}$, by double Weyl differencing over an exact carry-branch decomposition and master identity, with a targeted third differencing for the exact level-2 wave pieces (Lemma 5.2).

The sequences arise as the itineraries of the Juggler map $J(n) = \lfloor \sqrt{n} \rfloor$ (n even), $\lfloor n^{3/2} \rfloor$ (n odd), whose finite exact theory is developed in a companion manuscript [22]. As a corollary, the class of starts carrying a uniform power-envelope descent certificate of length at most four has natural density $13/16$, and an unconditional counting argument shows that parity equidistribution at *all* depths would give density one to the set of starts with some finite descent certificate. Neither is a density of starts that reach 1, and no statement about the Juggler conjecture itself is claimed. The remaining obstacle is the level-3 kernel, where the weight scale $n^{27/16}$ exceeds n ; we state it as an open problem and record precisely what blocks every method of this paper.

1. Introduction

Equidistribution questions for a *single* floor of a smooth function — the Piatetski-Shapiro tradition — are classical and well developed. This paper concerns the *composition* of floors:

$$m = \lfloor n^{3/2} \rfloor, \quad v = \lfloor m^{3/2} \rfloor, \quad w = \lfloor m^{1/2} \rfloor, \quad z = \lfloor v^{3/2} \rfloor, \dots$$

and, specifically, the joint distribution of the *parities* along such chains. Fix a pattern (an *itinerary word*) $w_1 w_2 \dots w_d$ over the alphabet $\{E, O\}$, where an O applies $x \mapsto \lfloor x^{3/2} \rfloor$ and an E applies $x \mapsto \lfloor \sqrt{x} \rfloor$. The question is whether the 2^d parity classes of the resulting chain equidistribute over odd starts $n \leq N$, with a power saving. Odd starts have $w_1 = O$; the theorems of this paper concern the O -rooted words. (Even starts are a separate and easier problem: one square root drops the state to scale $\sqrt{N}$, and we make no claims about them.)

The difficulty is genuinely one of composition. Writing $\theta = \{n^{3/2}\}$, the naive expansion $m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4})$ leaves a sawtooth of *growing* amplitude $n^{3/4}$ inside the phase. No estimate for exponential

sums with smooth monomial phases applies; the phase is not smooth, and the sawtooth cannot be discarded. All results in the single-floor literature stop before this point (Section 2).

Two ideas carry the paper.

1. **Exact linearization** (Lemma 4.3). The identity $m^{3/2} = \frac{3}{2}mn^{3/4} - \frac{1}{2}n^{9/4} + E(n)$ with $0 \leq E(n) \leq \frac{1}{2}n^{-3/4}$ eliminates the inner floor *exactly*: the integer m enters linearly, multiplied by a smooth coefficient, and the only non-smooth object left is m itself, which is handled by differencing and by an exact gap-cell decomposition. The identity is a Taylor rewrite, not a discovery; what carries the paper is the package built on it — one-signed remainders at every level, gap cells, the carry-branch decomposition, and a kernel with weight $n^{9/8}$. Iterated along the chain, this is what makes depth ≥ 2 accessible.
2. **The kernel theorem** (Theorem 5.3). Every reorganization of the depth-4 pattern OOO^* funnels into one object, the exponential sum of the level-2 floor defect

$$K_c(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e(c(n) \{ \lfloor n^{3/2} \rfloor^{3/2} \}), \quad c \asymp kP^{9/8}.$$

We prove $K_c(P) \ll P^{1-1/96+\varepsilon}$, uniformly for $k \leq P^{1/24}$, by double Weyl differencing over an exact carry-branch decomposition and master identity (Lemma 5.1), with a targeted third differencing for the level-2 wave pieces (Lemma 5.2). This is the hardest result of the paper, and Section 5 is written at full length — every estimate displayed with its constant — so that it can be checked without reference to anything outside this manuscript.

With the kernel theorem, depth-4 parity equidistribution over odd starts is complete (Theorem 6.1): the eight O -rooted length-4 words each receive their expected share with a power saving.

The chains are not chosen at random: they are the itineraries of the Juggler map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd,} \end{cases}$$

introduced by Pickover [1] (OEIS A094683 [2]); universal arrival at 1 is open. The map is niche; the nested-floor sums must stand on their own, and the paper is organized so that they do. The exact finite-word calculus of J — the power envelope $J^{|w|}(n)^{2^{|w|}} \leq n^{3^{\#O(w)}}$, the defect identity, and a small-cycle census — is a companion manuscript [22]; here we use only the contraction criterion (Proposition 3.1), whose short induction is written out below. The dynamical payoff of the counting theorems is one *certified-descent density*: the set of starts guaranteed to drop below their starting value within four steps has natural density $13/16$ (Corollary 4.9), and equidistribution at all depths would give the set of starts with *some* finite descent certificate density one (Proposition 7.1). We state plainly what these corollaries are not: they are not densities of starts that reach 1, and they do not touch the Juggler conjecture, whose analogue of Terras’s almost-all theorem for Collatz [4, 5, 6] remains open; see Lagarias [3] for the Collatz survey.

Section 7 states the frontier precisely. One nesting deeper, the same kernel reappears with weight scale $n^{27/16} > n$ (Conjecture 7.3), and every method of this paper stops below that scale. What survives is a clean model problem — the amplitude-product sums $\sum e(A(t)\{B(t)\})$ with $1 \ll A' \ll A$ — for which a direct L^2 computation in the shift of the fractional argument gives square-root cancellation, up to a $\sqrt{\log}$ factor, for almost every shift (Proposition 7.4). That average says nothing about the single deterministic shift; the deterministic instance (Conjecture 7.5) is open, and we record in three sentences why the obvious shortcuts fail.

1.1 Verification and evidence conventions

Every theorem in this paper is an ordinary human proof from the stated classical inequalities; none is machine-checked, and no numerical computation is used as a step in any proof below. Every exact identity used — the parity bridge $\lfloor x \rfloor$ odd iff $\{x/2\} \geq 1/2$, the gap-cell identity, the double-gap identity, and the master identity of Lemma 5.1 — is printed in the text with its proof. A companion repository (Section 8) contains machine formalizations and numerical probes of several of these identities; nothing in this paper depends on it. Densities of certificate classes are never densities of starts that reach 1.

2. Related work

Single floors. The distribution of $\lfloor n^c \rfloor$ ($c > 1$, $c \notin \mathbb{N}$) in arithmetic progressions and among primes goes back to Piatetski–Shapiro [12]; Leitmann [13] treated general smooth $\lfloor f(n) \rfloor$, and the prime exponent for $\lfloor n^c \rfloor$ has a long refinement history, e.g. Rivat–Sargos [14] and Rivat–Wu [15]. All of these concern a single floor of a smooth function; the analytic core is an exponential sum with smooth monomial phase, estimated by van der Corput’s method in the form presented by Graham–Kolesnik [11].

Digital and parity properties of $\lfloor n^c \rfloor$. Mauduit–Rivat [16] established q -multiplicative equidistribution along $\lfloor n^c \rfloor$ for c in an explicit range, Morgenbesser [17] proved the analogous statements for the sum of digits of $\lfloor n^c \rfloor$ in residue classes, and Müllner–Spiegelhofer [23] proved normality of the Thue–Morse sequence along $\lfloor n^c \rfloor$ for $1 < c < 3/2$. All of these estimate sums of the shape $\sum e(f(\lfloor n^c \rfloor))$ where f is a *digital* (automatic or q -additive) function: the outer function is handled by its own carry structure, and the analytic core is again a smooth-phase sum over the single floor. Parity of $\lfloor n^{3/2} \rfloor$ — our depth-1 statement, Theorem 4.1 — is the simplest instance of this circle and is classical; we include the short proof to fix constants and notation. Our outer function is not digital: it is a second convex power, and its fractional defect enters later phases with polynomially growing amplitude, which is a different regime.

Compositions and generalized polynomials. Two adjacent theories compose floors and do not cover our sums. For compositions of *Beatty* sequences $\lfloor \alpha \lfloor \beta n \rfloor \rfloor$, the inner floor error is bounded and the composition is again a bounded-perturbation linear sequence, so the linear theory applies. Bergelson–Leibman’s generalized polynomials [7] — expressions built from polynomials by iterated floors, sums, and products — admit a complete distributional theory via nilmanifolds; but that theory is confined to maps generated by *polynomials*, and $n^{3/2}$ is not one: no nilsystem models $\{f(\lfloor g(n) \rfloor)\}$ for convex non-polynomial powers f, g , and the growing-amplitude sawtooth that drives Sections 4–5 has no analogue there. We know of no published estimate for $\sum e(\xi\{f(\lfloor g(n) \rfloor)\})$ -type sums, or for the fractional parts $\{f(\lfloor g(n) \rfloor)\}$, with f, g convex powers and growing ξ .

Arithmetic of Piatetski–Shapiro sequences. Baker, Banks, Brüdern, Shparlinski, and Weingartner [18] study squarefree values, prime factors, and Carmichael numbers along $\lfloor n^c \rfloor$; Glasscock [21] studies linear equations whose unknowns range over Piatetski–Shapiro sequences. These results intersect a Piatetski–Shapiro sequence with other arithmetic sets; none composes two floors.

Beatty sequences. For the linear case $\lfloor \alpha n \rfloor$ the discrepancy theory is governed by the continued fraction of α (three-distance/Ostrowski structure); see Abercrombie [19] and, for character sums along Beatty sequences, Banks–Shparlinski [20]. The linear case admits exact self-similar structure that the convex case $n^{3/2}$ does not.

What is not covered. We know of no published equidistribution or parity result for *nested* floor powers $\lfloor \lfloor n^c \rfloor^d \rfloor$ with $c, d > 1$, nor any treatment of the level-2 defect sums $K_c(P)$ of Section 5. The obstruction is structural, not incremental: after one floor the argument of the second floor is an integer sequence, not a smooth function, and its fractional defect enters later phases with growing amplitude. The individual devices of this paper are not new — the linearization of Lemma 4.3 is a Taylor rewrite, and the carry bookkeeping of Lemma 5.1 will be recognized by anyone who has differenced $\lfloor f(n) \rfloor$; what we believe is new is the package (one-signed remainders, gap cells, the master identity, and the kernel bound of Theorem 5.3) and the results it yields. A literature check was last refreshed in August 2026; we would welcome pointers to anything missed.

Dynamical context. For the Collatz map, Terras [4] and Everett [5] proved that almost every start has finite stopping time, and Tao [6] proved that almost all orbits attain almost bounded values; Lagarias [3] surveys the problem. These are methodological cousins — density statements through parity-word counting — and motivate Proposition 7.1; they prove nothing about J , whose branches are floor powers rather than affine maps.

Standard tools. The van der Corput estimates are used in the form presented by Graham–Kolesnik [11]; see also Iwaniec–Kowalski [9, Ch. 8]. The indicator expansions use Vaaler’s extremal functions [10], and the discrepancy bridge is the Erdős–Turán inequality in the form of Kuipers–Niederreiter [8].

3. Preliminaries

Throughout, n denotes an odd integer, $e(t) = e^{2\pi it}$, $\{x\}$ the fractional part, and $\psi(x) = (-1)^{\lfloor x \rfloor}$. On a dyadic block $n \in (P, 2P]$ we write $n \sim P$. For the chains,

$$X = n^{3/2}, \quad m = \lfloor X \rfloor, \quad \theta = X - m; \quad Y = m^{3/2}, \quad v = \lfloor Y \rfloor, \quad \theta_2 = Y - v;$$

and on even m , $U = m^{1/2}$, $w = \lfloor U \rfloor$, $\theta_w = U - w$. An itinerary word of depth d at n is the sequence of parities $\text{word}_d(n)$ of $n, J(n), \dots, J^{d-1}(n)$.

Proposition 3.1 (power-envelope contraction; companion [22]). If the word w is realized at $n \geq 2$ (the orbit's parities are the letters of w) and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Write $k = |w|$ and $m = J^k(n)$. We first prove the envelope $m^{2^k} \leq n^{3^{\#O(w)}}$, by induction on prefix length. The empty prefix is $n \leq n$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $x^{2^\ell} \leq n^{3^o}$, and the next letter is realized. If that letter is even, then $J(x)^2 \leq x$, so $J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{2^\ell} \leq n^{3^o}$. If it is odd, then $J(x)^2 \leq x^3$, so $J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{3 \cdot 2^\ell} = (x^{2^\ell})^3 \leq (n^{3^o})^3 = n^{3^{o+1}}$. This is the envelope at w . The exponent gap and $n \geq 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, hence $m^{2^k} < n^{2^k}$. Since $m \geq 1$, one has $m < n$. $\square$

The companion [22] develops the same envelope as a finite-word identity with an exact defect; only the contraction criterion is used below. The induction is recorded here so that the criterion does not depend on an unpublished text.

A word w with $3^{\#O(w)} < 2^{|w|}$ is *contracting*; realizing a contracting word is a *descent certificate* of length $|w|$. Contracting words used below: E , OE , and $OOEE$. (Longer contracting words exist — $OOOEE$ and $OOEOE$ are the shortest beyond these — but this paper makes no counting claims about them.) Every even start realizes E , and every odd start with even image realizes OE ; those two certificates cover all starts except the odd-to-odd class, which is where the counting problem lives.

Lemma 3.2 (parity bridge). $\lfloor x \rfloor$ is odd if and only if $\{x/2\} \geq \frac{1}{2}$.

Proof. Write $x = \lfloor x \rfloor + \{x\}$ and split by the parity of $\lfloor x \rfloor$. $\square$

Lemma 3.3 (van der Corput, second-derivative form [11]). Let f be twice differentiable on an interval of length M , with $\lambda \leq |f''| \leq \alpha\lambda$ for some $\alpha \geq 1$. Then

$$\left| \sum e(f) \right| \ll_\alpha M\lambda^{1/2} + \lambda^{-1/2},$$

the sum over the integers of the interval.

The Weyl A -process used in Theorems 4.4 and 5.3 and in Lemma 5.2 is the following form, applied to a unimodular sequence on the odd integers of a block of length $\asymp P$ (equivalently, in the r -variable of Lemma 3.10, with shift h in r equal to shift $2h$ in n). For $1 \leq H \leq P$,

$$\left| \sum a_n \right|^2 \leq \frac{2P^2}{H} + \frac{4P}{H} \sum_{1 \leq h < H} \left| \sum a_{n+2h} \overline{a_n} \right|.$$

This is the classical inequality $|\sum a_n|^2 \leq \frac{P+2H}{H} \sum_{|h| < H} (1 - \frac{|h|}{H}) \sum a_{n+h} \overline{a_n}$ with the even-shift restriction and with the weights $1 - |h|/H \leq 1$ absorbed in the absolute constants. Every later appeal to “the classical inequality” is this display.

Lemma 3.4 (Erdős–Turán [8]). The discrepancy of R points $x_j \in \mathbb{R}/\mathbb{Z}$ against an interval satisfies, for every $H \geq 1$,

$$D \ll \frac{R}{H} + \sum_{h=1}^H \frac{1}{h} \left| \sum_{j \leq R} e(hx_j) \right|.$$

Lemma 3.5 (Vaaler [10]). For every $J \geq 1$ there are trigonometric polynomials $V_J(t) = \sum_{0 < |q| \leq J} a_q e(qt)$ with $|a_q| \leq \min(1, \frac{2}{|q|})$ and a nonnegative trigonometric polynomial $\Delta_J \geq 0$ of degree J with constant term

and coefficients at most $1/(J+1)$, such that the period-2 square wave satisfies $\psi(x) = V_J(x/2) + O(\Delta_J(x/2))$. For products, $|\psi_1\psi_2 - V^{(1)}V^{(2)}| \leq \Delta^{(1)} + \Delta^{(2)} + \Delta^{(1)}\Delta^{(2)}$, and each error term is again a nonnegative trigonometric polynomial of the same shape.

The class indicators below are products of half-wave factors $\frac{1}{2}(1 \pm \psi)$ evaluated along *formal* chains — chains whose branches are dictated by the target word, not by the orbit. The next lemma is the exact identity that justifies this; it replaces any appeal to sampled verification.

Lemma 3.6 (branch consistency). Let $d \geq 1$ and let $w = w_1 \cdots w_d$ be a word with $w_1 = O$. For odd n , define the formal chain $x_1 = n$ and, for $1 \leq t < d$,

$$\xi_t = \begin{cases} x_t^{3/2}, & w_t = O, \\ x_t^{1/2}, & w_t = E, \end{cases} \quad x_{t+1} = \lfloor \xi_t \rfloor,$$

and set $\sigma_t = +1$ if $w_t = E$, $\sigma_t = -1$ if $w_t = O$. Then, for every odd n ,

$$\prod_{t=1}^{d-1} \frac{1}{2}(1 + \sigma_{t+1} \psi(\xi_t)) = [\text{word}_d(n) = w].$$

Proof. Induct on d . For $d = 1$ both sides are 1. Let Π_{d-1} be the product over $t \leq d-2$; by induction $\Pi_{d-1} = [\text{word}_{d-1}(n) = w_1 \cdots w_{d-1}]$. If $\Pi_{d-1} = 0$, both sides vanish, since every factor lies in $[0, 1]$. If $\Pi_{d-1} = 1$, then the orbit realizes $w_1 \cdots w_{d-1}$, and a second induction along the chain gives $x_t = J^{t-1}(n)$ for $t \leq d-1$: assuming $x_t = J^{t-1}(n)$, the true parity of x_t is w_t , so J applies exactly the w_t -branch and $J^t(n) = \lfloor \xi_t \rfloor = x_{t+1}$. Hence ξ_{d-1} is the true real image with $\lfloor \xi_{d-1} \rfloor = J^{d-1}(n)$, and the last factor $\frac{1}{2}(1 + \sigma_d \psi(\xi_{d-1}))$ equals 1 exactly when the parity of $J^{d-1}(n)$ is w_d , else 0. $\square$

Each $\psi(\xi_t)$ is defined for *every* odd n , which is what permits the unconditional Vaaler expansion of the product: off the cylinder, the vanishing of an earlier factor kills the term, and on the cylinder every factor computes the true letter.

The last three preliminaries are the remaining analytic tools of Section 5: a finite Fourier expansion for sawtooth phases whose coefficient may exceed 1; a second-derivative test for two-term monomial phases in which the curvature transition — the short set where the two curvatures cancel — receives a trivial bound; and a sublevel splitting for three-term monomial phases, again with the transition set destined for a trivial bound. No third- or fourth-derivative exponential-sum test is used anywhere in this paper: on the short cells of Section 5 those tests do not sum, and the transition sets are cheaper to measure than to oscillate over. A final remark fixes the parity reindexing once and for all.

Lemma 3.7 (finite Fourier expansion with a shifted window). Let $B \in \mathbb{R}$, and let T, J be integers with $J \geq 1$ and $T \geq 8(1 + |B|)$. There exist coefficients $(b_u)_{|u| \leq T}$ with

$$|b_u| \leq \min\left(2, \frac{1}{\pi|u+B|}\right) + \min\left(2, \frac{1}{\pi|u|}\right), \quad \sum_{|u| \leq T} |b_u| \leq 8 + 2 \log(2 + |B| + T),$$

and coefficients $(v_q)_{0 < |q| \leq J}$ with $|v_q| \leq \min(2, 2\pi|B|)/|q|$, such that for every $t \in \mathbb{R}$

$$\left| e(-B\{t\}) - \sum_{|u| \leq T} b_u e(ut) - \sum_{0 < |q| \leq J} v_q e(qt) \right| \leq \min(2, 2\pi|B|) \Delta_J(t) + \frac{8(1 + |B|)}{T},$$

where $\Delta_J \geq 0$ is a trigonometric polynomial of degree J with constant term and coefficients at most $1/(J+1)$.

Proof. Write $f(t) = e(-B\{t\})$ and $\sigma(t) = \frac{1}{2} - \{t\}$, the sawtooth with jump $+1$ at the integers. The jump of f at the integers is $1 - e(-B)$, so $g := f - (1 - e(-B))\sigma$ is continuous, 1-periodic, and piecewise C^1 . Since $|f| = 1$, $|\sigma| \leq \frac{1}{2}$, and $|1 - e(-B)| = 2|\sin \pi B| \leq \min(2, 2\pi|B|) \leq 2$, we have $|g| \leq 2$, hence $|\hat{g}(0)| \leq 2$. For $u \neq 0$, direct integration gives $\hat{f}(u) = (1 - e(-B))/(\pi i(u + B))$ when $u + B \neq 0$ and $\hat{\sigma}(u) = 1/(\pi i u)$, so

$$\hat{g}(u) = \frac{1 - e(-B)}{2\pi i} \left(\frac{1}{u + B} - \frac{1}{u} \right) = -\frac{(1 - e(-B))B}{2\pi i u(u + B)}.$$

Two bounds follow. If $\pi|u+B| \geq \frac{1}{2}$, the first form gives $|\hat{g}(u)| \leq \frac{2}{2\pi}(\frac{1}{|u+B|} + \frac{1}{|u|}) = \frac{1}{\pi|u+B|} + \frac{1}{\pi|u|}$; if $\pi|u+B| < \frac{1}{2}$, then $|\hat{g}(u)| \leq |\hat{f}(u)| + |1 - e(-B)| |\hat{\sigma}(u)| \leq 1 + \frac{1}{\pi|u|} \leq 2 + \frac{1}{\pi|u|}$. In both cases (and when $u+B=0$, where $|\hat{f}(u)| \leq 1$ directly)

$$|\hat{g}(u)| \leq \min\left(2, \frac{1}{\pi|u+B|}\right) + \min\left(2, \frac{1}{\pi|u|}\right),$$

and summing this over $|u| \leq T$ — at most four terms hit each of the two caps, and the harmonic tails integrate to logarithms — gives $\sum_{|u| \leq T} |\hat{g}(u)| \leq 8 + 2 \log(2 + |B| + T)$. The second form gives, for $|u| > T \geq 8(1 + |B|)$ (so $|u+B| \geq |u| - |B| \geq |u|/2$),

$$|\hat{g}(u)| \leq \frac{2|B|}{2\pi} \cdot \frac{2}{u^2} = \frac{2|B|}{\pi u^2}, \quad \sum_{|u| > T} |\hat{g}(u)| \leq \frac{4|B|}{\pi T} \leq \frac{8(1 + |B|)}{T}.$$

The full series $\sum_u |\hat{g}(u)|$ therefore converges, and since g is continuous its Fourier series converges to g everywhere; take $b_u = \hat{g}(u)$ for $|u| \leq T$ and charge the tail to the flat error. For the sawtooth part, Vaaler's theorem [10] provides a trigonometric polynomial $V_J^*(t) = \sum_{0 < |q| \leq J} a_q^* e(qt)$ with $|a_q^*| \leq 1/|q|$ and $|\sigma(t) - V_J^*(t)| \leq \Delta_J(t)$ with Δ_J as stated; multiply by $1 - e(-B)$ and set $v_q = (1 - e(-B))a_q^*$. $\square$

When the lemma is applied with $t = G(n)$ along a block, the Δ_J -term is a nonnegative majorant: its sum over the block is $P/(J+1)$ plus J -truncated mode sums of G with coefficients $\leq 1/(J+1)$, and the flat term contributes $8(1 + |B|)P/T$. Both costs are displayed at each application below.

Lemma 3.8 (two-term monomial test with a trivial transition bound). Let $E \subset \mathbb{Q} \cap [-4, 4]$ be a fixed finite set disjoint from $\{0, 1, 2, 3\}$, let $\alpha \neq \beta$ lie in E , let $I \subseteq (P, 2P]$, and let

$$f(n) = a n^\alpha + b n^\beta + g(n), \quad M := \max(|a|P^{\alpha-2}, |b|P^{\beta-2}) \leq 1,$$

where the perturbation satisfies, for every $n \in I$,

$$|g''(n)| \leq \rho(|a|n^{\alpha-2} + |b|n^{\beta-2}), \quad |g'''(n)| \leq \rho(|a|n^{\alpha-3} + |b|n^{\beta-3}),$$

with $\rho \leq \rho_0(E)$ sufficiently small in terms of E alone. Then

$$\left| \sum_{n \in I} e(f(n)) \right| \ll_E |I| M^{1/2} + M^{-1/2} + (P/M)^{1/3}.$$

(If $M \leq P^{-2}$ the third term exceeds $|I|$ and the bound is trivial; the content is the range $M > P^{-2}$.)

Proof. Write $A(n) = a\alpha(\alpha-1)n^{\alpha-2}$ and $B(n) = b\beta(\beta-1)n^{\beta-2}$, so that

$$f'' = A + B + g'', \quad n f''' = (\alpha-2)A + (\beta-2)B + n g'''.$$

On the block, $|A(n)| \asymp_E |a|P^{\alpha-2}$ and $|B(n)| \asymp_E |b|P^{\beta-2}$. The ratio $|A/B| = |a\alpha(\alpha-1)/(b\beta(\beta-1))| n^{\alpha-\beta}$ is strictly monotone in n , so for any threshold $K = K(E) \geq 3$ the block splits into at most three consecutive intervals: I_1 where $|A/B| \geq K$, I_2 where $|A/B| \leq 1/K$, and a middle I_0 . (If a or b vanishes, only I_1 or I_2 is present and the argument shortens accordingly.)

On I_1 : $|f''| \geq (1 - \frac{1}{K})|A| - |g''| \geq \frac{1}{2}|A| \gg_E M$ -scale, with $\sup |f''|/\inf |f''| \leq C(E)$; Lemma 3.3 gives $\ll_E |I| M^{1/2} + M^{-1/2}$. Symmetrically on I_2 .

On I_0 , $|A| \asymp_K |B|$. If A and B have the same sign there, $|f''| \geq |A| + |B| - |g''| \geq |A|$ and Lemma 3.3 applies as before. If they have opposite signs, write $B = -sA$ with $s = s(n) \in [1/K, K]$, monotone in n . Then

$$f'' = A(1-s) + g'', \quad n f''' = A((\alpha-2) - s(\beta-2)) + n g'''.$$

The two affine functions $s \mapsto 1-s$ and $s \mapsto (\alpha-2) - s(\beta-2)$ have distinct zeros: $s = 1$ and $s = (\alpha-2)/(\beta-2) \neq 1$ (here $\alpha \neq \beta$ and $\beta \neq 2$ are used). Hence

$$c_6(E) := \min_{s > 0} \max(|1-s|, |(\alpha-2) - s(\beta-2)|) > 0.$$

Since $s(n)$ is monotone, the set where $|1 - s(n)| \geq c_6$ is the complement of a single interval, so I_0 splits into at most three consecutive intervals: two on which $|f''| \geq \frac{1}{2}c_6|A|$ (after absorbing g by $\rho_0(E) \leq c_6/8$ — Lemma 3.3 as before — and one transition interval J on which $|nf''| \geq \frac{1}{2}c_6|A|$ throughout, i.e. $|f''| \geq c_8(E)M/P$ in normalized scale. On J the second derivative is strictly monotone. Let $V \in (0, M]$ be a parameter: the set $\{n \in J : |f''(n)| < V\}$ is a single interval of length at most $2V/(c_8M/P)$, which receives the trivial bound, and its at most two flanking intervals carry $|f''| \geq V$ of a single sign with $\sup |f''| \leq C(E)M$, so Lemma 3.3 (applied at the piece's own infimum) gives $\ll_E |I|M^{1/2} + V^{-1/2}$ for each. Choosing $V = (c_8M/P)^{2/3}$ equalizes $V^{-1/2}$ and the trivial cost at $\ll_E (P/M)^{1/3}$. Summing the $O_E(1)$ contributions proves the lemma. $\square$

The proof's only content is that the second and third derivatives of a two-term monomial phase cannot both be small on the same subinterval — the linear system with matrix $\begin{pmatrix} 1 & 1 \\ \alpha-2 & \beta-2 \end{pmatrix}$ is invertible — so the transition set where the second-derivative test fails is short, and *measuring* it costs less than any attempt to oscillate over it: crucially, the transition cost $(P/M)^{1/3}$ carries no factor $|I|$, so it sums over many short cells. All applications in Section 5 use exponent pairs from $E = \{\frac{3}{4}, \frac{5}{4}, \frac{11}{8}, \frac{3}{2}, \frac{15}{8}\}$. One corner of Section 5 (three curvature scales meeting on a zero-offset branch) needs the three-term analogue, which we state directly as a sublevel bound: there the transition set is measured once against a global smooth model, and only the second-derivative test is ever applied.

Lemma 3.9 (three-term monomial sublevel splitting). Let E be as in Lemma 3.8 and let $\alpha, \beta, \gamma \in E$ be pairwise distinct, $I \subseteq (P, 2P]$, and

$$f(n) = an^\alpha + bn^\beta + cn^\gamma + g(n), \quad S := \max(|a|P^{\alpha-2}, |b|P^{\beta-2}, |c|P^{\gamma-2}) > 0,$$

with g satisfying the analogues of the perturbation bounds of Lemma 3.8 for the second, third, and *fourth* derivatives, at some $\rho \leq \rho_0(E)$. There are constants $c_7(E) > 0$ and $C(E)$ such that for every V with $0 < V \leq c_7S/2$:

- (i) the sublevel set $\Omega_V = \{n \in I : |f''(n)| \leq V\}$ is a union of at most $C(E)$ intervals of total length

$$|\Omega_V| \leq C(E) \left(\frac{PV}{S} + P \left(\frac{V}{S} \right)^{1/2} \right);$$

- (ii) $I \setminus \Omega_V$ is a union of at most $C(E)$ intervals, on each of which f'' is single-signed with $V \leq |f''| \leq C(E)S$. If one of a, b, c vanishes the claim reduces to Lemma 3.8; if two vanish, to Lemma 3.3. The application in Theorem 5.3, Step 5b, uses the three-term statement when a window mode is present and the two-term reduction when $w = 0$.

Proof. Write A, B, C for the three curvature terms $a\alpha(\alpha-1)n^{\alpha-2}$ etc., so that

$$f'' = A + B + C + g'', \quad nf''' = (\alpha-2)A + (\beta-2)B + (\gamma-2)C + ng''',$$

$$n^2f'''' = (\alpha-2)(\alpha-3)A + (\beta-2)(\beta-3)B + (\gamma-2)(\gamma-3)C + n^2g''''.$$

The coefficient matrix is the Vandermonde-type matrix in the distinct values $\alpha-2, \beta-2, \gamma-2$ (rows 1, x , $x(x-1)$), hence invertible with inverse bounded in terms of E : at every point of I ,

$$\max(|f''|, |nf''', |n^2f''''|) \geq c_7(E) \max(|A|, |B|, |C|) \geq c_7(E) \tilde{S}, \quad \tilde{S} \asymp_E S,$$

after absorbing g by $\rho_0 \leq c_7/8$ (zero coefficients only shorten the argument). The ratios $A/B, B/C, A/C$ are monotone, so I splits into $O_E(1)$ consecutive intervals on each of which one fixed derivative order $r \in \{2, 3, 4\}$ is good at full scale throughout. On an $r = 2$ piece, Ω_V is empty (as $V \leq c_7S/2$). On an $r = 3$ piece, $|f''| \geq c_7S/(2P)$ makes f'' strictly monotone, so Ω_V meets it in a single interval of length $\leq 4PV/(c_7S)$. On an $r = 4$ piece, $|f''''| \geq c_7S/(4P^2)$ makes f'''' strictly monotone, so f'' has at most one interior critical point and Ω_V meets the piece in at most two intervals; if $[x, y] \subseteq \Omega_V$, the second-difference identity $f''(x) - 2f''(\frac{x+y}{2}) + f''(y) = f''''(\xi)(y-x)^2/4$ forces $(y-x)^2 \leq 16V \cdot 4P^2/(c_7S)$, i.e. $y-x \leq 8P(V/(c_7S))^{1/2}$. Summing the $O_E(1)$ pieces gives (i). For (ii): on the complement, f'' is continuous with $|f''| \geq V$, hence single-signed on each maximal interval, and $|f''| \leq |A| + |B| + |C| + |g''| \leq C(E)S$. For

the triple $(\frac{5}{4}, \frac{11}{8}, \frac{3}{2})$ of Theorem 5.3, Step 5b, the ℓ^∞ operator norm of the inverse is 288, and one may take $c_7 = 1/288$. $\square$

The lemma is a measure statement, not an exponential-sum estimate: in its one application (Theorem 5.3, Step 5b) the sublevel set receives the trivial bound and the complement the second-derivative test, with the parameter V chosen there.

Lemma 3.10 (parity reindexing). All counting sums below run over odd n . The substitution $n = 2r + 1$ maps a block interval of length L onto an interval of length $L/2$ and a phase f to $f^*(r) = f(2r + 1)$ with $(f^*)^{(k)}(r) = 2^k f^{(k)}(2r + 1)$. Consequently:

- (a) any two terms of a composite phase have the same curvature *ratios* and the same *signs* in the r -variable as in the n -variable, so every dominance margin and single-sign check displayed below is invariant under the substitution;
- (b) if $\lambda \leq |f''| \leq \alpha\lambda$ on an interval of length L , then Lemma 3.3 applied in the r -variable (length $L/2$, curvature window $[4\lambda, 4\alpha\lambda]$) gives

$$\left| \sum_{n \in I, n \text{ odd}} e(f(n)) \right| \ll_\alpha \frac{L}{2} (4\lambda)^{1/2} + (4\lambda)^{-1/2} \leq L\lambda^{1/2} + \lambda^{-1/2} :$$

the n -variable display dominates the true reindexed bound, and the same check for Lemma 3.8 replaces its conclusion by a smaller quantity;

- (c) sublevel and transition sets (Lemmas 3.8–3.9) are measured in the n -line, and a set of total length Λ contains at most $\Lambda/2 + C(E)$ odd integers, so trivial bounds displayed in n -length dominate.

Every application of Lemmas 3.3 and 3.7–3.9 in Sections 4–6 is to be read through this substitution; no displayed constant below needs adjustment.

4. Exact linearization and depths one to four

For odd n write $s(n) = \psi(n^{3/2}) = (-1)^m$ and $S_O(N) = \sum_{n \leq N, n \text{ odd}} s(n)$. Let $M(N)$ be the number of odd $n \leq N$ and $\text{OO}(N) = \#\{n \leq N : n \text{ odd}, J(n) \text{ odd}\}$, so $S_O(N) = M(N) - 2\text{OO}(N)$. By Lemma 3.2, with $g(r) = \frac{1}{2}(2r + 1)^{3/2}$, $s(2r + 1) = -1$ iff $\{g(r)\} \geq \frac{1}{2}$, and $S_O(N)$ is (twice) an interval discrepancy of the sequence $\{g(r)\}$.

Theorem 4.1 (depth one; classical). $|S_O(N)| \ll N^{5/6}$.

Proof. This is the standard van der Corput estimate for the monomial g , included for completeness. $g''(r) = \frac{3}{2}(2r + 1)^{-1/2}$ is positive and decreasing. Partition $1 \leq r < R$ into dyadic blocks $[M, \min(2M, R))$; on a block $g'' \asymp M^{-1/2}$, so for the h -th mode $f = hg$ has $\lambda \asymp hM^{-1/2}$ and Lemma 3.3 gives $|\sum_{r \asymp M} e(hg(r))| \ll h^{1/2}M^{3/4} + h^{-1/2}M^{1/4}$. Lemma 3.4 with $H = \lfloor M^{1/6} \rfloor$ bounds the block's discrepancy by $\ll M^{5/6}$, and the dyadic sum is $O(R^{5/6}) = O(N^{5/6})$. $\square$

Corollary 4.2 (density of the odd-to-odd class). $|\text{OO}(N) - N/4| \ll N^{5/6}$. Equivalently, the starts covered by the uniform one- or two-step certificates (E and OE) have natural density $3/4$.

This is a density of a certificate class, not of starts that reach 1. Theorem 4.1 concerns consecutive source intervals; it supplies no transfer theorem for orbit samples or sparse image sets, so it cannot by itself control the odd-to-odd dynamics after the first step. The rest of the paper does not transfer it; it attacks the nested sums directly, by removing the inner floors exactly before any analytic step.

Lemma 4.3 (exact linearization and gap cells). (i) For every odd $n \geq 3$,

$$m^{3/2} = \frac{3}{2} m n^{3/4} - \frac{1}{2} n^{9/4} + E(n), \quad 0 \leq E(n) \leq \frac{3}{8} (n^{3/2} - 1)^{-1/2} \leq \frac{1}{2} n^{-3/4}.$$

(ii) For $h \geq 1$ and odd n , let $g(n) = m(n + 2h) - m(n)$ and $\delta(n) = (n + 2h)^{3/2} - n^{3/2}$. Then

$$g(n) = \lfloor \delta(n) \rfloor + \kappa(n), \quad \kappa(n) = [\{n^{3/2}\} \geq 1 - \{\delta(n)\}] \in \{0, 1\}.$$

Proof. (i) Let $f(t) = (X-t)^{3/2}$ on $[0, \theta]$, so $f(\theta) = m^{3/2}$. Then $f'(t) = -\frac{3}{2}(X-t)^{1/2}$ and $f''(t) = \frac{3}{8}(X-t)^{-1/2}$. Taylor with Lagrange remainder at 0 gives $f(\theta) = f(0) + f'(0)\theta + \frac{1}{2}f''(\xi)\theta^2 = X^{3/2} - \frac{3}{2}X^{1/2}\theta + \frac{3}{16}(X-\xi)^{-1/2}\theta^2$ for some $\xi \in (0, \theta)$. Substituting $\theta = X - m$ in the linear term yields $-\frac{1}{2}X^{3/2} + \frac{3}{2}mX^{1/2} = -\frac{1}{2}n^{9/4} + \frac{3}{2}mn^{3/4}$. Since $\theta \in [0, 1)$ and $f'' > 0$, the remainder $E(n)$ lies in $[0, \frac{3}{16}(X-1)^{-1/2}]$. The coarser bound $\frac{3}{8}(X-1)^{-1/2}$ used in the statement absorbs the missing factor of 2 against $\theta^2 \leq 1$, and $(X-1)^{-1/2} \leq \frac{4}{3}X^{-1/2}$ already for $n \geq 3$, whence $E(n) \leq \frac{1}{2}n^{-3/4}$. (ii) $g = \lfloor X + \delta \rfloor - \lfloor X \rfloor = \lfloor \delta \rfloor + \lfloor \{X\} + \{\delta\} \rfloor$, and the last floor is 1 precisely when $\{X\} + \{\delta\} \geq 1$. $\square$

The point of (i): the non-smooth integer m enters the phase *linearly*, multiplied by a smooth coefficient; no fractional part of growing amplitude survives. The naive expansion $m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4})$ instead leaves the sawtooth θ with the growing amplitude $n^{3/4}$, which defeats the van der Corput method directly. For (ii), since δ is smooth and increasing with $\delta' \asymp hn^{-1/2}$, the level sets of $\lfloor \delta \rfloor$ partition a dyadic block $n \in (P, 2P]$ into $O(hP^{1/2})$ cells of length $\asymp P^{1/2}/h$ on which the integer part of the gap is constant.

Theorem 4.4 (nested parity discrepancy). For every $\varepsilon > 0$ and every $(a, b) \in \{0, 1\}^2 \setminus \{(0, 0)\}$,

$$\left| \sum_{n \leq N, n \text{ odd}} \psi(n^{3/2})^a \psi(m(n)^{3/2})^b \right| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the four joint parity classes of $(m, \lfloor m^{3/2} \rfloor)$ on odd $n \leq N$ has cardinality $\frac{N}{8} + O(N^{23/24+\varepsilon})$; in particular the odd-to-odd cylinder refines one letter deeper than Theorem 4.1.

Proof. Work on dyadic blocks $n \in (P, 2P]$, odd, with truncations $J_1 = J_2 = P^{1/24}$ (wave modes), $H = P^{1/12}$ (differencing), and $R = P^{1/4}$ (cell modes). All derivative tests are read through the parity reindexing of Lemma 3.10; every constant below already dominates the reindexed bound.

Step 1 (wave expansion). By Lemma 3.5 applied to each present factor, $\psi_1^a \psi_2^b$ differs from the product of the truncated polynomials $V^{(1)}V^{(2)}$ by at most $\Delta^{(1)} + \Delta^{(2)} + \Delta^{(1)}\Delta^{(2)}$, and each error layer is a non-negative trigonometric polynomial of degree J_1 resp. J_2 with constant term and coefficients $\leq 1/(J+1)$. Summed over the block, a majorant layer costs its constant term $\leq P/(2(J_1+1)) \leq P^{23/24}$ plus mode sums $\sum_{0 < r \leq 2J_1} (J_1+1)^{-1} |\sum_n e(\frac{r}{2}n^{3/2})|$: each inner sum has $|(\frac{r}{2}X)''| \in [0.26, 0.75]rP^{-1/2}$, single sign, so Lemma 3.3 gives $\leq 1.4r^{1/2}P^{3/4} + 2P^{1/4}$, and the weighted total is $\leq 2.4J_1^{1/2}P^{3/4} + 2P^{1/4} \leq 3P^{19/24}$. The theorem is thus reduced to bounds, uniform in the mode pair, for

$$S_{i,j}(P) = \sum_{n \in (P, 2P], n \text{ odd}} e(\frac{i}{2}n^{3/2} + \frac{j}{2}m^{3/2}), \quad |i| \leq 2J_1, \quad 1 \leq |j| \leq 2J_2,$$

with mode weights of total mass $O(\log^2 P)$; the $j = 0$ sums are the depth-1 sums of Theorem 4.1. Replacing the summand by its conjugate we may take $j \geq 1$.

Step 2 (linearization). By Lemma 4.3(i), replacing $\frac{j}{2}m^{3/2}$ by $\frac{3j}{4}mn^{3/4} - \frac{j}{4}n^{9/4}$ changes each summand by a phase of modulus $\leq \frac{j}{2}E(n) \leq \frac{j}{4}n^{-3/4}$, hence changes $S_{i,j}$ by at most $2\pi \frac{j}{4} \sum_{n \sim P} n^{-3/4} \leq 2jP^{1/4} \leq 4P^{7/24}$. Call the linearized phase Φ .

Step 3 (Weyl differencing). For $H = P^{1/12}$, the A -process recorded after Lemma 3.3 gives

$$|S_{i,j}|^2 \leq \frac{2P^2}{H} + \frac{4P}{H} \sum_{1 \leq h < H} |T_h|, \quad T_h = \sum_{\substack{n, n+2h \in (P, 2P] \\ n \text{ odd}}} e(\Phi(n+2h) - \Phi(n)).$$

Step 4 (the exact differenced phase). Write $\Delta f = f(\cdot+2h) - f(\cdot)$, $g(n) = m(n+2h) - m(n)$, and $m = n^{3/2} - \theta_n$. Exactly,

$$\Phi(n+2h) - \Phi(n) = \frac{3j}{4}g(n)(n+2h)^{3/4} + \frac{j}{2}A_h(n) - B_h(n)\theta_n + \frac{j}{2}\Delta(n^{3/2}),$$

with $A_h = \frac{3}{2}n^{3/2}\Delta(n^{3/4}) - \frac{1}{2}\Delta(n^{9/4})$ and $B_h = \frac{3j}{4}\Delta(n^{3/4})$. Taylor with third-order remainders gives, for $n \sim P$ and $h \leq H$,

$$\Delta(n^{3/4}) = \frac{3}{2}hn^{-1/4} - \frac{3}{8}h^2n^{-5/4} + O(h^3P^{-9/4}), \quad \Delta(n^{9/4}) = \frac{9}{2}hn^{5/4} + \frac{45}{8}h^2n^{1/4} + O(h^3P^{-3/4}),$$

so the two $n^{5/4}$ -scale contributions of A_h cancel exactly at leading order (both equal $\frac{9}{4}hn^{5/4}$) and

$$A_h = -\frac{27}{8}h^2n^{1/4}(1 + O(hP^{-1})), \quad |(\frac{j}{2}A_h)''| \leq 0.32jh^2P^{-7/4},$$

$$B_h \in [0.95, 1.13]jhP^{-1/4}, \quad |(\frac{j}{2}\Delta(n^{3/2}))''| = \frac{j}{2}2h|X'''(\xi)| \leq 0.38ihP^{-3/2}.$$

The sawtooth term is *discarded*: $|e(-B_h\theta_n) - 1| \leq 2\pi B_h \leq 7.1jhP^{-1/4}$ per term, so deleting it changes T_h by at most $3.6jhP^{3/4}$. This cost is charged in Step 7; it is admissible only because $jh \leq 2P^{1/24+1/12} = 2P^{1/8}$, which is the reason the truncation J_2 and the differencing length H cannot be enlarged in this argument.

Step 5 (cells and the exact shift device). Split T_h over the level sets of $[\delta]$, $\delta(n) = (n+2h)^{3/2} - n^{3/2}$: by (the argument of) Lemma 4.3(ii) and $\delta' \in (1.5, 2.2)hP^{-1/2}$, at most $1.5hP^{1/2} + 1$ cells of length in $[\frac{2}{3}, 0.95]P^{1/2}/h$. On a cell $g = G + \kappa$ with $G = [\delta]$ frozen and $\kappa(n) = [\{n^{3/2}\} \geq \rho(n)]$, $\rho = 1 - \{\delta(n)\}$ smooth and monotone there. Vaaler-expand κ (one sawtooth layer, Lemma 3.5) with modes $e(rn^{3/2})$, $0 < |r| \leq R$, weights $\leq 1/|r|$, majorant cost $\leq P/(R+1) + (R\text{-mode sums at coefficients } \leq 1/(R+1)) \leq P^{3/4} + P^{7/8}$. The moving endpoint costs nothing: since $\rho(n) = 1 + G - \delta(n)$, the coefficient's n -dependent piece contributes the exact smooth phase $r\delta(n)$, and $e(rn^{3/2} + r\delta(n)) = e(r(n+2h)^{3/2})$ up to a unimodular constant, so every r -term falls into one of the two smooth families $e(rn^{3/2})$, $e(r(n+2h)^{3/2})$; the remaining in-cell coefficient variation is exactly 1 per cell and one Abel summation absorbs it at a factor $\leq 2+2\pi$.

Step 6 (second-derivative test per cell). After Steps 4–5 the phase on a cell is $\frac{3j}{4}G(n+2h)^{3/4} + \frac{j}{2}A_h + \frac{j}{2}\Delta(n^{3/2})$ plus, for the r -pieces, $rn^{3/2}$ or $r(n+2h)^{3/2}$.

For $r = 0$: the main curvature is the single monomial term

$$\left(\frac{3j}{4}G(n+2h)^{3/4}\right)'' = -\frac{9j}{64}G(n+2h)^{-5/4} \in -[0.15, 0.68]jhP^{-3/4},$$

using $G \in (3hP^{1/2}-1, 4.3hP^{1/2})$ from (the mean value bound for) δ and $(n+2h)^{-5/4} \in (2^{-5/4}, 1]P^{-5/4}$. Its competitors are dominated at displayed margins: $|(\frac{j}{2}A_h)''|/0.15jhP^{-3/4} \leq 2.2hP^{-1} \leq 2.2P^{-11/12}$ and $|(\frac{j}{2}\Delta(n^{3/2}))''|/0.15jhP^{-3/4} \leq 2.5(i/j)P^{-3/4} \leq 5P^{1/24-3/4}$, so the composite second derivative is single-signed with in-cell ratio ≤ 4.6 . Lemma 3.3 per cell (through Lemma 3.10) and summation over cells give

$$\sum_{\text{cells}} (\ell_i \lambda^{1/2} + \lambda^{-1/2}) \leq 0.83(jh)^{1/2}P^{5/8} + 3.9(h/j)^{1/2}P^{7/8}.$$

For $r \neq 0$: the mode curvature $|(rX)''| \geq 0.53|r|P^{-1/2}$ dominates the full cell curvature $\leq 0.68jhP^{-3/4}$ at ratio $\geq 0.78|r|P^{1/4}/(jh) \geq 0.39P^{1/8} \geq 4$ for $P \geq P_0$ (here $jh \leq 2P^{1/8}$ is used again), so the composite keeps the mode's sign and, up to a factor $\frac{3}{4}$, its size. Lemma 3.3 per cell and the $1/|r|$ weights give

$$\sum_{0 < |r| \leq R} \frac{1}{|r|} \left(1.3|r|^{1/2}P^{3/4} + 1.5hP^{1/2} \cdot 1.6|r|^{-1/2}P^{1/4}\right) \leq 5.2R^{1/2}P^{3/4} + 13hP^{3/4} \leq 6P^{7/8} + 13P^{5/6}.$$

Step 7 (totals and assembly). Collecting Steps 4–6,

$$|T_h| \leq C((jh)^{1/2}P^{5/8} + (h/j)^{1/2}P^{7/8} + P^{7/8} + jhP^{3/4}) \log P \leq C'P^{7/8}(1 + h^{1/2}) \log P$$

uniformly in the modes (for the last step: $(jh)^{1/2}P^{5/8} \leq 2^{1/2}P^{1/16+5/8} \leq P^{3/4}$ and $jhP^{3/4} \leq 2h^{1/2} \cdot (jh^{1/2})P^{3/4} \leq 2h^{1/2}P^{1/24+1/24+3/4} \leq 2h^{1/2}P^{7/8}$, using $j \leq 2P^{1/24}$, $h^{1/2} \leq P^{1/24}$). Hence

$$|S_{i,j}|^2 \leq \frac{2P^2}{H} + \frac{4P}{H} \cdot C' \log P \sum_{h < H} P^{7/8}(1 + h^{1/2}) \leq 2P^{23/12} + 6C'P^{15/8}H^{1/2} \log P \ll P^{23/12+\varepsilon}$$

at $H = P^{1/12}$, so $|S_{i,j}| \ll P^{23/24+\varepsilon}$. Summing mode weights ($O(\log^2 P)$), the majorant and truncation costs of Steps 1 and 5 ($\leq P^{23/24} + 3P^{19/24} + P^{7/8}$ per layer), and dyadic blocks gives the theorem. $\square$

The exponent $23/24$ is deliberately unoptimized; every stage has slack. The proof above is the undecorated instance of Lemma 5.2(i) below, which reruns the same six stages with three classes of decorations riding along; the reader who has checked this proof has checked the skeleton of that one.

Proposition 4.5 (OE-branch third letter). For $a \in \{0, 1\}$,

$$\left| \sum_{n \leq N, n \text{ odd}} ((-1)^m)^a \psi(m^{1/2}) \right| \ll_\varepsilon N^{7/8+\varepsilon},$$

and consequently $\#\text{OEO}(N)$, $\#\text{OEE}(N) = N/8 + O(N^{7/8+\varepsilon})$. Together with Theorem 4.4 this makes depth 3 complete: each of OOO , OOE , OEO , OEE has density $1/8$ with a power saving.

Proof. Taylor expansion of $(X - \theta)^{1/2}$ with both correction terms one-signed gives

$$m^{1/2} = n^{3/4} + D_1(n), \quad -\frac{1}{2}X^{-1/2} - \frac{1}{8}(X-1)^{-3/2} \leq D_1 \leq 0.$$

D_1 is decaying, so replacing $\frac{1}{2}m^{1/2}$ by $\frac{1}{2}n^{3/4}$ inside a Vaaler mode costs $\ll l \sum_{n \sim P} n^{-3/4} \ll lP^{1/4}$, absorbable at $l \leq 2P^{1/24}$. The mode sums are then pure: $\sum_{n \sim P} e(\frac{i}{2}n^{3/2} + \frac{j}{2}n^{3/4})$ with $l \neq 0$. For $i \neq 0$, $\varphi'' \asymp |i|P^{-1/2}$ is single-signed and Lemma 3.3 gives $\ll i^{1/2}P^{3/4} + i^{-1/2}P^{1/4}$; for $i = 0$, $\varphi'' \asymp lP^{-5/4}$ gives $\ll l^{1/2}P^{3/8} + l^{-1/2}P^{5/8}$. Vaaler majorants, truncation tails, and dyadic assembly are as in Theorem 4.4, and every bound is $\ll P^{7/8}$ after summing mode weights. Branch consistency is Lemma 3.6: the $(1 + (-1)^m)/2$ factor vanishes exactly on odd m , where the true chain takes the $3/2$ -power branch. $\square$

Lemma 4.6 (fourth-letter linearization). For every odd $n \geq 3$,

$$v^{1/2} = n^{9/8} + D(n), \quad -\frac{3}{4}n^{-3/8} - n^{-9/8} \leq D(n) \leq 0.$$

Proof. Two applications of the Lemma 4.3(i) pattern. With $f(t) = (Y - t)^{1/2}$ on $[0, \theta_2]$: $v^{1/2} = m^{3/4} - \frac{1}{2}\theta_2 m^{-3/4} - E_2$, $0 \leq E_2 \leq \frac{1}{8}(Y-1)^{-3/2}$. With $f(t) = (X - t)^{3/4}$ on $[0, \theta]$: $m^{3/4} = n^{9/8} - \frac{3}{4}\theta n^{-3/8} - E_3$, $0 \leq E_3 \leq \frac{3}{32}(X-1)^{-5/4}$. Every term is nonpositive after the leading one, and $\frac{1}{2}m^{-3/4} + E_3 + E_2 \leq n^{-9/8}$ for $n \geq 3$. $\square$

Unlike the depth-2 layer, *every* non-smooth term now has decaying amplitude: the cumulative phase cost of replacing $\frac{k}{2}v^{1/2}$ by the smooth $\frac{k}{2}n^{9/8}$ over a dyadic block is $\ll kP^{5/8}$, negligible against the target.

Theorem 4.7 (triple parity discrepancy). For every $\varepsilon > 0$ and every $(a, b, c) \in \{0, 1\}^3 \setminus \{(0, 0, 0)\}$,

$$\left| \sum_{n \leq N, n \text{ odd}} \psi(n^{3/2})^a \psi(m^{3/2})^b \psi(v^{1/2})^c \right| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the eight sign classes of $(\psi(n^{3/2}), \psi(m^{3/2}), \psi(v^{1/2}))$ on odd $n \leq N$ has cardinality $\frac{N}{16} + O(N^{23/24+\varepsilon})$.

Proof. The $c = 0$ cases are Theorem 4.4 and Theorem 4.1. For $c = 1$, wave-expand all present sign factors as in Step 1 of Theorem 4.4, with truncations $J_1 = J_2 = J_3 = P^{1/24}$, and bound

$$S_{i,j,k}(P) = \sum_{n \in (P, 2P], n \text{ odd}} e(\frac{i}{2}n^{3/2} + \frac{j}{2}m^{3/2} + \frac{k}{2}v^{1/2})$$

uniformly for $|i| \leq 2J_1$, $|j| \leq 2J_2$, $1 \leq |k| \leq 2J_3$. By Lemma 4.6, replacing $\frac{k}{2}v^{1/2}$ by $\frac{k}{2}n^{9/8}$ costs $\ll |k|P^{5/8} \leq P^{2/3}$, before any differencing.

If $j = 0$, the remaining sum is a single smooth exponential sum with phase $\frac{i}{2}n^{3/2} + \frac{k}{2}n^{9/8}$. Both curvature terms are positive for positive modes, and for mixed signs with $|i| \geq 1$ the first dominates by $P^{3/8-1/24}$; Lemma 3.3 gives $\ll P^{5/8}$ for $i = 0$ and $\ll |i|^{1/2}P^{3/4} + P^{1/4}$ otherwise, both $\ll P^{23/24}$.

If $j \neq 0$, the phase is exactly the Theorem 4.4 phase plus the smooth passenger $\frac{k}{2}n^{9/8}$. Steps 2–6 run verbatim; the passenger modifies the smooth part of the differenced phase by $\frac{k}{2}[(n+2h)^{9/8} - n^{9/8}]$, whose

second derivative $\ll |k|hP^{-15/8}$ is smaller than the retained cell curvature $jhP^{-3/4}$ and the r -mode curvature $|r|P^{-1/2}$ by powers of P . Every sign-dominance check of Theorem 4.4 holds with these margins. $\square$

Theorem 4.8 (the OE splits).**

$$\#\text{OEOE}(N), \#\text{OEOO}(N) = \frac{N}{16} + O(N^{7/8+\varepsilon}), \quad \#\text{OEEE}(N), \#\text{OEEO}(N) = \frac{N}{16} + O(N^{13/16+\varepsilon}).$$

Together with Theorems 4.4 and 4.7 this proves every depth-4 itinerary word class except $OOO*$.

Proof. On the OE branch m is even and $w = \lfloor U \rfloor$, $U = m^{1/2}$. The w -level linearization is Lemma 4.3(i) verbatim with base U : since $Um^{1/4} = m^{3/4}$ exactly,

$$w^{3/2} = m^{3/4} - \frac{3}{2} m^{1/4} \theta_w + E, \quad 0 \leq E \leq \frac{3}{8} (U-1)^{-1/2},$$

and $m^{3/4} = n^{9/8}$ up to one-signed corrections of scale $n^{-3/8}$. For a Vaaler mode $k \neq 0$, the phase is therefore $\varphi_1(n) - B(n)\theta_w(n)$ up to absorbable errors, with $\varphi_1 = \frac{i}{2}n^{3/2} + \frac{j}{2}n^{3/4} + \frac{k}{2}n^{9/8}$ and $B = \frac{3k}{4}n^{3/8}(1 + O(n^{-3/2}))$: a *single growing sawtooth* of amplitude $\asymp kn^{3/8}$ riding $\theta_w = \{m^{1/2}\}$, whose underlying integer increments once every $\asymp \frac{4}{3}n^{1/4}$ steps. Truncations: $J_1 = J_2 = J_3 = P^{1/8}$.

Drift-1 intervals. $B' \asymp kn^{-5/8}$, so B drifts by at most 1 on intervals I of length $L_0 = P^{5/8}/k$; there are $\asymp kP^{3/8}$ of them. On I , expand $e(-B\{U\}) = \sum_r a_r(B)e(rU)$ (Fourier in U , Vaaler truncation $|r+B| \leq T = P^{5/16}$); the coefficients satisfy $|a_r(B)| \leq \min(1, |r+B|^{-1})$ with window mass $O(\log T)$, and their n -dependence through $B(n)$ has total variation $O(1)$ per interval, removed by one partial summation per mode.

Mode sums. Smoothing $rU \rightarrow rn^{3/4}$ costs $kP^{5/8}$ in total. Writing $r = -B_0 + t$, $|t| \leq T$, the curvature of the mode phase is

$$\varphi'' = \frac{27k}{128}n^{-7/8} - \frac{3}{16}(t + \frac{j}{2})n^{-5/4} + \frac{3i}{8}n^{-1/2}.$$

For $i \neq 0$ the last term dominates and Lemma 3.3 gives $\ll i^{1/2}P^{3/4} + i^{-1/2}kP^{5/8}$ after summing intervals. For $i = 0$, $(T + J_2)P^{-5/4} \ll kP^{-7/8}$ because $T = P^{5/16} \ll kP^{3/8}$ for every $k \geq 1$, so $\lambda \asymp kn^{-7/8}$ is single-signed; Lemma 3.3 per interval and summation over $\asymp kP^{3/8}$ intervals give $S_k \ll k^{1/2}P^{13/16+\varepsilon}$. With Vaaler weights $1/k$, $\sum_{k \leq J_3} k^{-1/2}P^{13/16} = J_3^{1/2}P^{13/16} = P^{7/8}$, balanced against the truncation error $P/J_3 = P^{7/8}$ at $J_3 = P^{1/8}$. Total $\ll P^{7/8+\varepsilon}$.

OEE branch. The fourth letter needs $\psi(w^{1/2})$, and $w^{1/2} = U^{1/2} - \frac{1}{2}\theta_w U^{-1/2} - \frac{1}{8}\theta_w^2(U - \xi)^{-3/2}$ has *decaying* amplitudes only, with $U^{1/2} = m^{1/4} \rightarrow n^{3/8}$ likewise. The pure modes $e(\frac{l}{2}n^{3/8})$ have $\lambda \asymp lP^{-13/8}$, giving $l^{1/2}P^{3/16} + l^{-1/2}P^{13/16}$; the mixed cases sit in the dominance hierarchy $iP^{-1/2} \gg jP^{-5/4} \gg lP^{-13/8}$ with all three second derivatives of the same sign. Total $\ll N^{13/16+\varepsilon}$. $\square$

Corollary 4.9 (certified-descent density 13/16). The three uniform certificate classes

$$E, \quad OE, \quad OOOE$$

are disjoint, and

$$\begin{aligned} \#\{n \leq N : \text{word}(n) \text{ has prefix } E\} &= \lfloor \frac{N}{2} \rfloor, \\ |\#\{n \leq N : \text{word}(n) \text{ has prefix } OE\} - \frac{N}{4}| &\ll N^{5/6}, \\ |\#\{n \leq N : \text{word}(n) \text{ has prefix } OOOE\} - \frac{N}{16}| &\ll_\varepsilon N^{23/24+\varepsilon}. \end{aligned}$$

Hence the class of starts with a certified descent within four steps has cardinality

$$\frac{N}{2} + \frac{N}{4} + \frac{N}{16} + O(N^{23/24+\varepsilon}) = \frac{13N}{16} + O(N^{23/24+\varepsilon}).$$

No other depth- ≤ 4 word class is used. In particular this is not a census of E -rooted words of length ≥ 2 , and it is not a census of every O -rooted word of length four (the classes $OOO*$ are Theorem 6.1, and they are non-contracting at this depth).

Proof. Every even start realizes E , so the first count is elementary. The OE count is Corollary 4.2. For $OOEE$, Lemma 3.6 gives $\#OOEE = \frac{1}{8} \sum_{n \leq N \text{ odd}} (1 - \psi_1)(1 + \psi_2)(1 + \psi_3)$ with $\psi_1 = \psi(n^{3/2})$, $\psi_2 = \psi(m^{3/2})$, $\psi_3 = \psi(v^{1/2})$; expanding gives the main term $N/16$ and seven sign sums bounded by Theorem 4.7. Every $OOEE$ start descends within four steps by Proposition 3.1: $3^2 < 2^4$. The three classes are disjoint because they are distinct prefixes. $\square$

The density $13/16$ is the exact ceiling of this one-growing-layer machinery: a word contracts iff $3^o < 2^\ell$, the method so far proves letters at positions 1–3 of any word plus further letters along even branches only, and the contracting minimal words with all odd letters at positions ≤ 2 are exactly E , OE , and $OOEE$. Completing depth 4 requires the OOO^* split — a second growing layer, where the fourth-letter phase coefficient $W \asymp kn^{9/8}$ crosses integers within single steps and no drift-1 interval exists. Sections 5 and 6 close that split. (The OOO^* words are non-contracting at depth 4, so the certified density remains $13/16$.)

5. The kernel theorem

Every reorganization of the OOO^* phase funnels into one object: for smooth weights c of scale $kP^{9/8}$ on $n \sim P$,

$$K_c(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e(c(n) \{ \lfloor n^{3/2} \rfloor^{3/2} \}).$$

This section proves the kernel bound at full length: every estimate that the proof uses is displayed, with its dependence on the mode k , the differencing shifts h_1, h_2, h_3 , the branch offset j , and the expansion frequencies made explicit, and with numerical constants displayed at each of the sign-critical dominance checks. Two structural facts shape the proof and are worth stating in advance. First, the hardest pieces of the decomposition are exact level-2 waves $e(qY)$, possibly riding a frozen floor; a frozen-coefficient model $e(sX)$ with $s \asymp qP^{3/4}$ would silently discard the sawtooth $-\frac{3}{2}qX^{1/2}\theta$ of the same amplitude hidden in $Y = (X - \theta)^{3/2}$, so the waves are treated exactly — by the same exact linearization that proves Theorem 4.4 — and the honest bound is $q^{-1/6}P^{23/24+\varepsilon}$ (Lemma 5.2): the wave is a depth-2 object and receives exactly depth-2 strength, which is what pins the kernel saving at $\delta = \frac{1}{96}$. Second, the offset-branch pieces of Step 5 carry a factor $(k|j|)^{1/2}$, and $(k|j|)^{1/2}P^{15/16} \leq P^{23/24}$ exactly on the standing range $k \leq P^{1/24}$: the two bottlenecks meet at the endpoint, which is why the theorem is uniform in k on that range and why the range cannot be enlarged by this assembly.

The next lemma collects the exact reductions behind the treatment: the kernel phase is the level-2 local floor defect; the second difference of the level-2 integer obeys an exact floor identity; on an explicit carry-branch decomposition that second difference is a smooth function with a frozen floor; and the doubly differenced kernel phase decomposes exactly into four bounded pieces — the master identity — with no growing smooth part left to estimate.

Lemma 5.1 (level-2 defect, double gap, branch freeze, and the master identity). Fix shifts $d_1 = 2h_1$, $d_2 = 2h_2$ and write Δ_1 , Δ_2 , $\Delta\Delta$ for the corresponding difference operators in n , and $W = \Delta_1 Y$, $W' = \Delta_2 Y$.

(i) For odd $n \geq 3$,

$$\frac{3}{4}v^{1/2}\theta_2 = \frac{1}{2}(m^{9/4} - v^{3/2}) - R, \quad 0 \leq R \leq \frac{3}{16}v^{-1/2}.$$

Hence for $c = \frac{3k}{4}v^{1/2}$ the kernel phase $c\theta_2$ equals $\frac{k}{2}(Y^{3/2} - v^{3/2})$ up to $kR \ll kP^{-9/8}$ per term: the kernel is the exponential sum of the level-2 local floor defect.

(ii) Exactly, with $g_2 = \Delta_1 v$,

$$\Delta_2 g_2 = \lfloor \Delta\Delta Y \rfloor + \kappa'' + \Delta_2 \kappa_2, \quad \kappa_2 = [\theta_2 \geq 1 - \{W\}], \quad \kappa'' = [\{W\} \geq 1 - \{\Delta\Delta Y\}],$$

and every carry is a difference of unit sawtooths: $\lfloor \{A\} + \{B\} \rfloor \geq 1 = \{A\} + \{B\} - \{A+B\}$ for all reals A, B (both sides equal $\{A\} + \{B\} - \{A+B\}$: the left side because $\{A\} + \{B\} \in [0, 2)$ and $\{A+B\} = \{A\} + \{B\} - \lfloor \{A\} + \{B\} \rfloor$).

- (iii) Write $b_1 = \lfloor \Delta_1 X \rfloor$, $b_2 = \lfloor \Delta_2 X \rfloor$, $b_{12} = \lfloor \Delta_{12} X \rfloor$ (shift $d_1 + d_2$) — each constant on runs of length $\asymp P^{1/2}/h$ — and let $\kappa = (\kappa_1, \kappa_2, \kappa_{12}) \in \{0, 1\}^3$ be the level-1 carries, so that the shifted values of m are $m + \beta_1$, $m + \beta_2$, $m + \beta_{12}$ with $\beta_i = b_i + \kappa_i$. On each b -run intersection and carry branch,

$$\Delta \Delta Y = F_\kappa(m), \quad F_\kappa(m) = (m + \beta_{12})^{3/2} - (m + \beta_1)^{3/2} - (m + \beta_2)^{3/2} + m^{3/2}$$

exactly. The net offset $j = \beta_{12} - \beta_1 - \beta_2$ satisfies $|j| \leq 3$ for $h_1 h_2 \leq P^{1/2}/3$, and F splits exactly into an offset term and a genuine second difference: for some ξ_1 between 0 and j and some $\xi_2 \in (0, \beta_1 + \beta_2)$,

$$F_\kappa(m) = \frac{3}{2} j (m + \beta_1 + \beta_2 + \xi_1)^{1/2} + \frac{3}{4} \beta_1 \beta_2 (m + \xi_2)^{-1/2}.$$

Consequently, with $G := F_\kappa \circ X$,

$$\frac{3}{2} |j| P^{3/4} \leq \left| \frac{3}{2} j (\cdot)^{1/2} \right| \leq 2.6 |j| P^{3/4}, \quad 1.4 h_1 h_2 P^{1/4} \leq \frac{3}{4} \beta_1 \beta_2 (\cdot)^{-1/2} \leq 15 h_1 h_2 P^{1/4},$$

$$|G'(n)| \leq 2 |j| P^{-1/4} + 20 h_1 h_2 P^{-3/4} < 1, \quad |G''(n)| \leq 2 |j| P^{-5/4} + 25 h_1 h_2 P^{-7/4},$$

so $\lfloor G(n) \rfloor$ is constant on runs of length $\geq \frac{1}{22} \min(P^{1/4}/(|j|+1), P^{3/4}/(h_1 h_2))$. The branch indicator is a finite union of arcs in the single variable θ with slowly moving endpoints (drifts $\leq 1.5 h P^{-1/2}$ per step): exactly the moving-endpoint pattern of Step 4 of Theorem 4.4, absorbed by the same exact shift device at $O(\log P)$ mode-mass cost.

- (iv) (*Master identity.*) With $\kappa_2 = [\theta_2 \geq 1 - \{W\}]$, $\kappa'_2 = [\theta_2 \geq 1 - \{W'\}]$ the level-2 carries at the two shifts, and κ'' , $\Delta_2 \kappa_2$ as in (ii), the doubly differenced kernel phase decomposes exactly:

$$\Delta \Delta (c \theta_2) = (\Delta \Delta c) \theta_2 + (\Delta_2 c)(n + d_1) (\{W\} - \kappa_2) + (\Delta_1 c)(n + d_2) (\{W'\} - \kappa'_2) + c_{11} (\{\Delta \Delta Y\} - \kappa'' - \Delta_2 \kappa_2),$$

where c_{11} is the doubly shifted weight. Every bracket is bounded by 2 in absolute value: no unbounded smooth part survives the differencing.

Proof. (i) Taylor of $(v + \theta_2)^{3/2}$ at v , with $Y^{3/2} = m^{9/4}$. (ii) The gap identity of Lemma 4.3(ii) applied twice — to Y at shift d_1 , then to the real sequence $n \mapsto W(n)$ at shift d_2 , with $\Delta_2 W = \Delta \Delta Y$; the sawtooth form of the carry is the displayed elementary identity. (iii) The corner identities are Lemma 4.3(ii) at the three shifts, and the offset bound follows from $|\Delta \Delta X| \leq 4 h_1 h_2 \sup |X''| = 3 h_1 h_2 P^{-1/2} < 1$. For the split, group $F = [(m + \beta_1 + \beta_2 + j)^{3/2} - (m + \beta_1 + \beta_2)^{3/2}] + [(m + \beta_1 + \beta_2)^{3/2} - (m + \beta_1)^{3/2} - (m + \beta_2)^{3/2} + m^{3/2}]$: the first bracket is $\frac{3}{2} j (\cdot)^{1/2}$ by the mean value theorem, the second is $\beta_1 \beta_2 f''(m + \xi_2)$ for $f = x^{3/2}$ by the double mean value theorem. The displayed numerical bounds use $m \in (P^{3/2} - 1, 2^{3/2} P^{3/2}]$, $\beta_i \in [3 h_i P^{1/2} - 1, 3 \sqrt{2} h_i P^{1/2} + 1]$ (from $\Delta_i X = 2 h_i X'(\xi)$ and (E1) below), and $G' = F'(X) X'$, $G'' = F''(X) X'^2 + F'(X) X''$ with $F' = \frac{3}{4} j (\cdot)^{-1/2} - \frac{3}{8} \beta_1 \beta_2 (\cdot)^{-3/2}$, $F'' = -\frac{3}{8} j (\cdot)^{-3/2} + \frac{9}{16} \beta_1 \beta_2 (\cdot)^{-5/2}$. (iv) By (ii) and the definitions: $\Delta_1 \theta_2 = \Delta_1 Y - \Delta_1 v = W - (\{W\} + \kappa_2) = \{W\} - \kappa_2$ (Lemma 4.3(ii) applied to Y), likewise $\Delta_2 \theta_2 = \{W'\} - \kappa'_2$, and $\Delta \Delta \theta_2 = \Delta \Delta Y - \Delta_2 g_2 = \{\Delta \Delta Y\} - \kappa'' - \Delta_2 \kappa_2$ by (ii). Substitute these into the exact product rule

$$\Delta \Delta (c f) = c_{11} \Delta \Delta f + (\Delta_2 c)(n + d_1) \Delta_1 f + (\Delta_1 c)(n + d_2) \Delta_2 f + (\Delta \Delta c) f,$$

which is verified by expanding both sides over the four base points $n, n + d_1, n + d_2, n + d_1 + d_2$. $\square$

One warning, essential to the organization: $\lfloor \Delta \Delta Y \rfloor$ itself is *not* frozen. The level-1 carry toggles at essentially every step and shifts $\Delta \Delta Y$ by $\frac{3}{2} j_1 m^{1/2}$, a jump of size $\asymp P^{3/4}$. The branch decomposition of (iii), which carries the flicker inside the indicator over the eight carry branches, is therefore forced: no decomposition conditioning on the *values* of the level-1 gaps produces sets on which the second difference is smooth.

Standing estimates

All estimates of this section take place on a dyadic block $n \in (P, 2P]$, n odd, under the standing constraints

$$(C1) \quad k h_1 h_2 \leq P^{1/8}, \quad (C2) \quad h_1 h_2 \leq P^{1/2}/3, \quad (C3) \quad 1 \leq k \leq P^{1/24},$$

and every displayed constant below is valid for $P \geq P_0$ with an absolute P_0 . All sums run over odd n , and every derivative test below is read through the parity reindexing of Lemma 3.10: margins and signs are

invariant, and each displayed test bound dominates its reindexed value, so no constant needs adjustment. The base derivatives are

$$(E1) \quad X' = \frac{3}{2}n^{1/2} \in (1.5, 2.13] P^{1/2}, \quad X'' = \frac{3}{4}n^{-1/2} \in [0.53, 0.75] P^{-1/2}, \\ -X''' = \frac{3}{8}n^{-3/2} \in [0.13, 0.375] P^{-3/2}, \quad X'''' = \frac{9}{16}n^{-5/2} \leq 0.57 P^{-5/2},$$

and $m = X - \theta \in (P^{3/2} - 1, 2.83 P^{3/2}]$. Every gap is a mean value: for any shift $2h$,

$$(E2) \quad \Delta X = 2hX'(\xi) \in (3hP^{1/2}, 4.3hP^{1/2}), \quad \beta := \lfloor \Delta X \rfloor + \kappa \in (3hP^{1/2} - 1, 4.3hP^{1/2} + 1),$$

so that every level-1 gap integer at shift $2h$ is $\asymp hP^{1/2}$ with the displayed constants. For the monomial weight $c = \frac{3k}{4}n^{9/8}$ (the general c of Theorem 5.3 changes only the absolute constants),

$$(E3) \quad c' = \frac{27}{32}kn^{1/8}, \quad c'' = \frac{27}{256}kn^{-7/8}, \quad c''' = -\frac{189}{2048}kn^{-15/8}, \quad c'''' = \frac{2835}{16384}kn^{-23/8},$$

$$(E4) \quad \Delta_i c = 2h_i c'(\xi) \in (1.68, 1.85) kh_i P^{1/8}, \quad \Delta \Delta c = 4h_1 h_2 c''(\xi) \in (0.22, 0.43) kh_1 h_2 P^{-7/8}.$$

On the level-2 side, with β_i as in Lemma 5.1(iii),

$$(E5) \quad W = \Delta_1 Y \in (4.4, 11) h_1 P^{5/4}, \quad |(W \circ \text{cell})'| = |W'_1(m) X'| \in (2.2, 10.4) h_1 P^{1/4},$$

where $W_1(m) = (m + \beta_1)^{3/2} - m^{3/2}$, so $\{W\}$ is a *fast* sawtooth (its argument moves by $\gg 1$ per step), while θ , $\{\Delta_i X\}$, and $\{\Delta \Delta Y\}$ are slow. The branch bounds of Lemma 5.1(iii) give, per branch with offset j ,

$$(E6) \quad |(cF)''| = \frac{945}{512} k|j| n^{-1/8} (1 + O(P^{-1/4})) + O(kh_1 h_2 P^{-5/8}),$$

the two shapes being the composite monomials $kjn^{15/8}$ (offset branches) and $kh_1 h_2$ -scale $n^{11/8}$ (zero-offset), with sign product $\alpha(\alpha - 1)(\alpha + \beta - 2)(\alpha + \beta - 3) > 0$ at $\alpha = \frac{9}{8}$, $\beta \in \{\frac{3}{4}, \frac{1}{4}\}$. Finally, the run and cell inventories: level-1 gap cells at shift $2h$ number at most $1.5hP^{1/2} + 1$ with lengths in $[\frac{2}{3}, 0.95] P^{1/2}/h$, and the frozen-floor runs of $\lfloor F_\kappa(X) \rfloor$ number at most $22(|j| + 1)P^{3/4}$ by the derivative bound of Lemma 5.1(iii).

Throughout, “ A is dominated by B at margin $P^{-\rho}$ ” means $\sup |A|/\inf |B| \leq CP^{-\rho}$ with the displayed C : the composite second derivative then keeps the sign and, up to the factor $1 + CP^{-\rho}$, the size of B , so Lemma 3.3 applies at B ’s scale.

Every numerical margin in this section is claimed only for $P \geq P_0$, with an absolute but ineffective P_0 . The comparisons stratify. The Lemma 3.7 / Lemma 5.2 window hypotheses first hold together on the printed majorants at a moderate threshold (of size $3 \cdot 10^5$). The interpolant error of Step 5b is $O(P^{-5/6})$ with leading coefficient 0.11, and is not absorbed into a smaller constant. The comparison $V \leq c_7 S/2$ at $c_7 = 1/288$ for the printed triple of Step 5b forces P_0 to be large (of size 10^{24}): until then a three-term zero of Φ'' can keep the sublevel Ω_V of length $\Theta(P)$ on a dyadic block, and the printed length $P^{89/96}$ is the bound of Lemma 3.9, not a claim that the sublevel is short at small P . The constants are not asserted to be sharp.

The proof of the kernel theorem classifies the differenced pieces into four classes; three are handled by standard tests, and the fourth — an exact level-2 wave $e(qY)$, possibly riding a frozen floor — is the genuinely new difficulty. We isolate it as a standalone lemma with its own differencing, so that it can be checked independently.

Lemma 5.2 (level-2 waves: the mixed-piece bound). Assume (C1)–(C3), write $\mathcal{D} = \{0, d_1, d_2, d_1 + d_2\}$, and call a *decoration* any sum ρ of at most nine terms of the classes

- (D1) $q' \Delta_{2h} \Delta_{2h'} Y(n + d')$ with $|q'| \leq 4P^{1/24}$, $1 \leq h' \leq 2P^{1/24}$, $d' \in \mathcal{D}$;
- (D2) $-\Delta_{2h}(c \lfloor F_\kappa(X) \rfloor)(n)$, with F_κ a branch function of Lemma 5.1(iii), offset $|j| \leq 3$;
- (D3) a smooth φ with $|\varphi''| \leq 3kh_1 h_2 P^{-5/8}$ and $|\varphi'''| \leq 3kh_1 h_2 P^{-13/8}$;

here $2h$ is the shift of part (i), and in part (ii) the decorations are read after the differencing there. Then, for sums over $n \in (P, 2P]$, n odd:

(i) (*differenced wave*) for all integers $u, h \geq 1$ with $h \leq P^{1/8}$, $uh \leq P^{1/2}$, and $d \in \mathcal{D}$,

$$V := \left| \sum_n e(u \Delta_{2h} Y(n+d) + \rho(n)) \right| \ll \left((uh)^{1/2} P^{5/8} + (h/u)^{1/2} P^{7/8} + P^{7/8} \right) P^\varepsilon;$$

(ii) (*waves*) for all integer coefficients $(q_d)_{d \in \mathcal{D}}$ with $|q_d| \leq 4P^{1/24}$ and $t := \sum_{d \in \mathcal{D}} q_d \neq 0$, all $\varepsilon_0 \in \{0, 1\}$, and φ of class (D3),

$$U := \left| \sum_n e \left(\sum_{d \in \mathcal{D}} q_d Y(n+d) - \varepsilon_0 c(n) [F_\kappa(X(n))] + \varphi(n) \right) \right| \ll |t|^{-1/6} P^{23/24+\varepsilon}.$$

Part (ii) is the mixed-piece bound proper: level-2 waves $e(qY)$, possibly riding a frozen floor. Part (i) is the engine that (ii)'s targeted differencing feeds, and is also used directly for the $\{W\}$ -content of the kernel proof. The exponent $\frac{23}{24}$ is not an accident: written exactly, the wave $e(qY)$ is the depth-2 object of Theorem 4.4, and part (ii) recovers exactly the depth-2 strength — no more. A model of U as $e(sX)$ with a frozen real coefficient $s \asymp tP^{3/4}$ would discard the sawtooth $-\frac{3}{2}tX^{1/2}\theta$ of amplitude $\asymp tP^{3/4}$ inside $tY = t(X - \theta)^{3/2}$; no such shortcut is taken anywhere below.

Proof of (ii) from (i). Replacing the summand by its conjugate we may take $t \geq 1$. Fix $e_1 \in \mathcal{D}$ with $q_{e_1} \neq 0$ and write, exactly (telescoping onto the base e_1),

$$\sum_{d \in \mathcal{D}} q_d Y(n+d) = t Y(n+e_1) + \sum_{d \neq e_1} q_d \Delta_{|d-e_1|} Y(n + \min(d, e_1)),$$

and the second sum is a set of at most three decoration seeds with shifts $|d - e_1|/2 \leq h_1 + h_2 \leq 2P^{1/24}$. Weyl differencing at shift $2h_3$, $1 \leq h_3 \leq H_3 := \lceil t^{1/3} P^{1/12} \rceil$, by the A -process recorded after Lemma 3.3:

$$|U|^2 \leq \frac{P^2}{H_3} + \frac{2P}{H_3} \sum_{h_3=1}^{H_3} |V_{h_3}|, \quad V_{h_3} = \sum_n e(t \Delta_{2h_3} Y(n+e_1) + \rho_{h_3}(n)),$$

where ρ_{h_3} collects $q_d \Delta_{2h_3} \Delta_{|d-e_1|} Y$ (class (D1) with $h' = |d-e_1|/2 \leq 2P^{1/24}$), $-\varepsilon_0 \Delta_{2h_3} (c[F(X)])$ (class (D2)), and $\Delta_{2h_3} \varphi$ (class (D3): $|(\Delta_{2h_3} \varphi)''| \leq 2h_3 \sup |\varphi'''| \leq 6kh_1 h_2 h_3 P^{-13/8} \leq 3kh_1 h_2 P^{-5/8}$ since $h_3 \leq P^{1/4}$). The hypothesis of (i) holds: $th_3 \leq t \cdot 2t^{1/3} P^{1/12} = 2t^{4/3} P^{1/12}$. With $|q_d| \leq 4P^{1/24}$ one has $t \leq 16P^{1/24}$, so $2t^{4/3} P^{1/12} \leq 81P^{5/36} \leq P^{1/2}$ for $P \geq P_0$, and $H_3 \leq 3P^{7/72} \leq P^{1/4}$. Part (i) gives

$$|U|^2 \leq \frac{P^2}{H_3} + \left(4t^{1/2} H_3^{1/2} P^{13/8} + 4t^{-1/2} H_3^{1/2} P^{15/8} + 4P^{15/8} \right) P^\varepsilon,$$

and at $H_3 = \lceil t^{1/3} P^{1/12} \rceil$ the four terms are, in order, $\leq t^{-1/3} P^{23/12}$; $\leq 6t^{2/3} P^{5/3} = 6(tP^{-1/4}) t^{-1/3} P^{23/12} \leq 96P^{1/24-1/4} \cdot t^{-1/3} P^{23/12}$; $\leq 6t^{-1/3} P^{1/24} P^{15/8} = 6t^{-1/3} P^{23/12}$; and $\leq 4(t^{1/3} P^{-1/24}) t^{-1/3} P^{23/12} \leq 11P^{1/72-1/24} t^{-1/3} P^{23/12}$. Hence $|U|^2 \ll t^{-1/3} P^{23/12+\varepsilon}$, which is (ii).

Proof of (i). Six stages; write $\nu = n + d$ (a shift of the block by at most $d_1 + d_2 \leq P^{1/2}$, harmless in every estimate) and $\Delta = \Delta_{2h}$.

Stage 1 (the m -linear form). By Lemma 4.3(i) at the bases ν and $\nu+2h$,

$$u \Delta Y = u A_h(\nu) + \frac{3u}{2} g(\nu) (\nu+2h)^{3/4} - \frac{3u}{2} \theta_\nu \Delta(\nu^{3/4}) + u \Delta E,$$

with $g = m(\nu+2h) - m(\nu)$, $|u \Delta E| \leq uP^{-3/4}$ (total cost over the block $\leq 2\pi u P^{1/4} \leq 2\pi P^{3/4}$), and

$$A_h(\nu) = \frac{3}{2} \nu^{3/2} \Delta(\nu^{3/4}) - \frac{1}{2} \Delta(\nu^{9/4}) = -\frac{27}{8} h^2 \nu^{1/4} (1 + O(hP^{-1})) :$$

the two $\nu^{5/4}$ -scale contributions cancel exactly at leading order (both equal $\frac{9}{4} h \nu^{5/4}$), leaving $|u A_h''| \leq 0.64 u h^2 P^{-7/4}$. The sawtooth coefficient is

$$B(\nu) := \frac{3u}{2} \Delta(\nu^{3/4}) = \frac{9}{4} u h \xi^{-1/4} \in (1.89, 2.25] u h P^{-1/4}.$$

Stage 2 (gap cells and the exact shift device). Split into the level sets of $\lfloor \delta_h \rfloor$, $\delta_h(\nu) = (\nu+2h)^{3/2} - \nu^{3/2}$: at most $1.5hP^{1/2} + 1$ cells of length in $[\frac{2}{3}, 0.95] P^{1/2}/h$ (E1). On a cell $g = G + \kappa$ with $G = \lfloor \delta_h \rfloor$ frozen and $\kappa = [\theta_\nu \geq 1 - \{\delta_h\}]$; Vaaler-expand κ at truncation $R_0 = P^{1/4}$ (majorant cost $\leq 4P/R_0 = 4P^{3/4}$). The moving endpoint costs nothing: as in Theorem 4.4, Step 4, the endpoint's smooth part contributes the exact phase $r\delta_h(\nu)$, and $e(r\nu^{3/2} + r\delta_h(\nu)) = e(r(\nu+2h)^{3/2})$, so every r -term falls into the two smooth families $e(r\nu^{3/2})$, $e(r(\nu+2h)^{3/2})$, $0 < |r| \leq R_0$, with weights $\leq 1/|r|$; the in-cell variation of the coefficient is exactly 1 and one Abel summation absorbs it at a factor $\leq 2+2\pi$.

Stage 3 (the θ -sawtooth). Two regimes. (s1) If $uh \leq P^{3/16}$ then $|B| \leq 2.25P^{-1/16} < \frac{1}{2}$ for $P \geq P_0$: Lemma 3.7 with $T = P^{1/2}$, $J = R_0$ (hypothesis $T \geq 8(1+|B|)$: $P^{1/2} \geq 9$ for $P \geq P_0$) expands $e(-B_0\{\nu^{3/2}\} \dots)$ per window (here one window: the total drift of B is $\leq 0.6uhP^{-1/4} \leq 0.6P^{-1/16} < 1$) into the same two mode families at coefficient factor $\min(2, 2\pi|B|) \leq 14P^{-1/16}$, mass $O(\log P)$, flat cost $\leq 12P^{1/2}$, majorant $\leq 14P^{-1/16} \cdot 4P^{3/4}$. (s2) If $P^{3/16} < uh \leq P^{1/2}$ then $|B| \leq 2.25P^{1/4}$ and B drifts by at most 1 on windows of length $\geq P^{5/4}/(0.6uh) \geq 1.2P^{3/4}$: at most $0.6P^{1/4} + 1$ windows. Per window Lemma 3.7 at the centre B_0 ($T = P^{1/2} \geq 8(1+2.25P^{1/4})$ for $P \geq P_0$, since $P^{1/4} \geq 19$; flat cost $\leq 8(1+2.25P^{1/4})P^{1/2} \leq 27P^{3/4}$ in total) produces modes $e(w\nu^{3/2})$ whose coefficients decay as $\min(2, \frac{1}{\pi|w+B_0|}) + \min(2, \frac{1}{\pi|w|})$; the window-boundary cost is, using $uh > P^{3/16}$, $\leq (0.6P^{1/4}+1) \cdot 1.83(0.30uh)^{-1/2}P^{3/8} \leq 2.1P^{1/4-3/32+3/8} = 2.1P^{17/32} \leq P^{5/8}$. Modes with $|w|$ in the collision window are treated in Stage 5.

Stage 4 (the main curvature, $r = w = 0$). On a cell the remaining phase is $\frac{3u}{2}G(\nu+2h)^{3/4} + uA_h +$ (smooth decorations), with second derivative

$$-\frac{9}{32}uG(\nu+2h)^{-5/4} \in -[0.30, 1.35]uhP^{-3/4} \quad (\text{E2}),$$

single-signed with in-cell ratio $\sup/\inf \leq 4.5$: the second derivative of $\frac{3u}{2}G(\nu+2h)^{3/4}$ is the single term $\frac{3u}{2}G \cdot \frac{3}{4} \cdot (-\frac{1}{4})(\nu+2h)^{-5/4}$, with G frozen and bounded by (E2). The competitor uA_h'' has ratio $\leq 0.64uh^2P^{-7/4}/(0.30uhP^{-3/4}) \leq 2.2hP^{-1} \leq 2.2P^{-3/4}$; decoration competitors are bounded in Stage 6. Lemma 3.3 per cell and summation give

$$\sum_{\text{cells}} \left(\ell_i \lambda^{1/2} + \lambda^{-1/2} \right) \leq 2.3(uh)^{1/2}P^{5/8} + 2.8(h/u)^{1/2}P^{7/8}.$$

Stage 5 (nonzero modes and collisions). A mode $w \neq 0$ (or $r \neq 0$; identical treatment with weight $1/|r|$) adds the phase $w\nu^{3/2}$ of curvature $\geq 0.53|w|P^{-1/2}$. If $0.53|w|P^{-1/2} \geq 4 \cdot 1.35uhP^{-3/4}$, i.e. $|w| \geq 10.2uhP^{-1/4}$, the mode curvature dominates at margin ≥ 4 : Lemma 3.3 over the full block gives $\leq 1.4|w|^{1/2}P^{3/4} + 1.4|w|^{-1/2}P^{1/4}$, and the coefficient-weighted sums are $\leq 3R_0^{1/2}P^{3/4} \log P = 3P^{7/8} \log P$ (families of Stage 2, regime (s1)) and $\leq CP^{7/8} \log P$ (regime (s2) tails). If $|w| \leq 0.1uhP^{-1/4}$ the main curvature dominates at margin ≥ 4 and the mode rides along at Stage 4's scale, weight summable to $O(\log P)$. In the remaining collision window $0.1uhP^{-1/4} \leq |w| \leq 10.2uhP^{-1/4}$ (nonempty only in regime (s2)), the two curvatures can cancel: there the phase is the two-term monomial $a\nu^{3/4} + w\nu^{3/2}$ with $a = \frac{3u}{2}G \asymp uhP^{1/2}$ plus perturbations already shown ρ_0 -small, and Lemma 3.8 with $(\alpha, \beta) = (\frac{3}{4}, \frac{3}{2})$ applies per window with $M \in [0.03, 11]uhP^{-3/4}$ (both curvature scales lie in this window on the collision band, and $M \leq 11P^{-1/4}$):

$$\leq C_E(|I_w|M^{1/2} + M^{-1/2} + (P/M)^{1/3}) \quad \text{per window.}$$

Summing over the at most $0.6P^{1/4}+1$ windows — the transition cost $(P/M)^{1/3}$ carries no window-length factor, which is why it sums — with the $O(\log P)$ collision-band coefficient mass:

$$\sum_w |I_w|M^{1/2} \leq 3.4(uh)^{1/2}P^{5/8}, \quad \sum_w M^{-1/2} \leq 4.5(uh)^{-1/2}P^{5/8} \leq 4.5P^{5/8},$$

$$\sum_w (P/M)^{1/3} \leq 0.77P^{1/4} \cdot 3.3(uh)^{-1/3}P^{7/12} = 2.5(uh)^{-1/3}P^{5/6} \leq 2.5P^{5/6-1/16} = 2.5P^{37/48},$$

the last using $uh > P^{3/16}$ in regime (s2). The collision-band total is $\leq C((uh)^{1/2}P^{5/8} + P^{5/6}) \log P \leq CP^{7/8} \log P$.

Stage 6 (decorations). Each class is dominated at a displayed margin against the Stage-4 curvature $\geq 0.30uhP^{-3/4}$.

- (D1). On the branch decomposition of Lemma 5.1(iii) at the shift pair (h_3 -role h, h'): the arcs are absorbed by the shift device (log mass); the b -run boundaries add $\leq 1.5(h+2h')P^{1/2} + 2$ cells, at boundary cost $\leq 1.5(h+2h')P^{1/2} \cdot 1.83(0.30uh)^{-1/2}P^{3/8} \leq 5.1(h/u)^{1/2}P^{7/8} + 10.3h'(uh)^{-1/2}P^{7/8}$ (the second term is what the H_3 -averaging of part (ii) absorbs: it contributes $\leq 2h't^{-2/3}P^{43/24} \leq P^{1/24-1/8}t^{-1/3}P^{23/12}$ there). The θ -coefficient of the decoration is $\leq |q'|(2|j'|P^{-1/4} + 20hh'P^{-3/4}) \leq 24P^{1/24-1/4} + 160P^{1/24+1/8+1/24-3/4} = 24P^{-5/24} + 160P^{-13/24} < 1$: sub-unit, expanded in the Stage-2 families. Its smooth curvature has ratio $\leq |q'|(2|j'|P^{-5/4} + 25hh'P^{-7/4})/(0.30uhP^{-3/4}) \leq 80P^{1/24-1/2} + 672P^{1/24+1/24-1} \leq P^{-1/4}$: dominated.
- (D2). Write $\Delta(c[F]) = (\Delta c)[F] + c_+\Delta[F]$ with $c_+ = c(\cdot + 2h)$.
 - (a) $(\Delta c)[F] = (\Delta c)F - (\Delta c)\{F\}$: the smooth part has curvature $\leq ((\Delta c)F)'' \leq 7kh|j|P^{-9/8} + 28khh_1h_2P^{-13/8}$, ratio to the main curvature $\leq 24k|j|P^{-3/8}/u \leq 72P^{1/24-3/8}$: dominated. The sawtooth part has coefficient $\Delta c \in (1.68, 1.85)khP^{1/8}$, handled per windows of drift ≤ 1 : at most $2khP^{1/8} + 1$ windows, at boundary cost $\leq 2khP^{1/8} \cdot 1.83(0.30uh)^{-1/2}P^{3/8} \leq 7k(h/u)^{1/2}P^{1/2}$. Per window, Lemma 3.7 at $T = P^{1/2}$ yields modes $e(q''(F \circ X))$ of curvature $\leq (2khP^{1/8} + P^{1/2}) \cdot 3|j|P^{-5/4} \leq 18P^{-3/4} \cdot P^{-1/16}$, ratio to the main curvature $\leq 60P^{-1/16}$: dominated. The flat cost is, per point, $8(1+B_0)/T$, so in total $\leq 8(1+1.85khP^{1/8})P^{1/2} \leq 8P^{1/2} + 15khP^{5/8} \leq 23P^{1/24+1/8+5/8} = 23P^{19/24} \leq P^{7/8}$, using $h \leq P^{1/8}$ and (C3).
 - (b) $c_+\Delta[F]$: by the gap identity applied to the sequence $F \circ X$, $\Delta[F] = [\Delta F] + \kappa_F$. ΔF has total drift $\leq P \cdot 2h \sup |(F \circ X)''| \leq P \cdot 2h(2|j|P^{-5/4} + 25h_1h_2P^{-7/4}) \leq 13hP^{-1/4} + 50h_1h_2P^{-3/4} < 1$, so $[\Delta F]$ takes at most two values, on at most two intervals; per value the phase $-q_0c_+$ ($|q_0| \leq 2+6hP^{-1/4} \leq 3$) has curvature $\leq 0.4kP^{-7/8}$, ratio $\leq 1.4kP^{-1/8}/(uh) \leq 1.4P^{1/24-1/8}$: dominated. The carry $\kappa_F = \{F\} + \{\Delta F\} - \{F + \Delta F\}$ is a sum of unit sawtooths in slow variables (all drifts < 1 per step): Lemma 3.5 at truncation $P^{1/8}$ (majorant $\leq 3 \cdot 4P^{7/8}$) gives modes $e(q''(F \circ X))$ -type of curvature $\leq P^{1/8} \cdot 3|j|P^{-5/4} \leq 9P^{-9/8}$, ratio $\leq 30P^{-3/8}$: dominated.
- (D3). Ratio $\leq 2h \sup |\varphi'''|/(0.30uhP^{-3/4}) \leq 20kh_1h_2P^{-7/8}/u \leq 20P^{1/8-7/8}$: dominated.

Totals. Stages 1–6 bound V by

$$C \left((uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8} + P^{7/8} + k(h/u)^{1/2}P^{1/2} \right) \log^3 P,$$

and the fourth term is $\leq P^{1/24} \cdot (h/u)^{1/2}P^{1/2}$, absorbed by the second. This is (i) up to the stated P^ε .

Costs collected. Each class contributes to V as follows, all times P^ε .

Class	Bound contributed to V	Absorbed by
Stage 1 remainder $u\Delta E$	$P^{3/4}$	$P^{7/8}$
Stage 2 majorant / shift device	$P^{3/4}$	$P^{7/8}$
Stage 3 (s1) sawtooth	$P^{3/4}$	$P^{7/8}$
Stage 3 (s2) windows / collision	$(uh)^{1/2}P^{5/8} + P^{7/8}$	(i)
Stage 4, $r = w = 0$	$(uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8}$	(i)
Stage 5, non-collision modes	$P^{7/8}$	(i)
(D1) curvature	dominated at $P^{-1/4}$	Stage 4
(D1) run boundaries	$(h/u)^{1/2}P^{7/8} + h'(uh)^{-1/2}P^{7/8}$	(i); H_3 -average
(D2)(a) smooth / sawtooth	dominated; $P^{7/8}$ flat	Stage 4; (i)
(D2)(b) gap / carry	dominated; $P^{7/8}$ majorant	Stage 4; (i)
(D3)	dominated at $P^{-3/4}$	Stage 4

No decoration class is used later unless it appears in this table. $\square$

The balance to keep in mind: with the trivial bound $|V_{h_3}| \leq P$, part (ii)'s differencing returns nothing beyond the choice of H_3 ; the content of the lemma is that the floor, coefficient, and carry bookkeeping of Stages 3–6 survives the targeted differencing with the Stage-4 curvature intact. What is *not* available is the frozen-coefficient shortcut: per frozen run the run length $P^{1/4}$ is shorter than $\lambda^{-1/2} \asymp (uh)^{-1/2}P^{3/8}$ for small

uh , and the run-boundary cost $(h/u)^{1/2}P^{7/8}$, summed against the H_3 -average, is exactly what pins the final exponent at $\frac{23}{24}$.

Theorem 5.3 (kernel cancellation). Let c be smooth on $(P, 2P]$ with $c^{(r)} \asymp kP^{9/8-r}$ for $r = 0, \dots, 4$, derivative signs following the monomial pattern (e.g. $c = \frac{3k}{4}n^{9/8}$), and $1 \leq k \leq P^{1/24}$. Then

$$K_c(P) \ll P^{1-1/96+\varepsilon},$$

uniformly in k .

Proof. Work on a dyadic block $n \sim P$, odd.

Step 1 (double Weyl differencing). Set $H_1 = P^{1/48}$, $H_2 = P^{1/24}$. The A -process recorded after Lemma 3.3, applied twice, gives

$$|K_c|^2 \leq \frac{2P^2}{H_1} + \frac{4P}{H_1} \sum_{h_1=1}^{H_1} |T_1(h_1)|, \quad |T_1|^2 \leq \frac{2P^2}{H_2} + \frac{4P}{H_2} \sum_{h_2=1}^{H_2} |T_2(h_1, h_2)|,$$

with $T_2 = \sum_n e(\varphi_2)$, $\varphi_2 = \Delta\Delta(c\theta_2)$. For all $h_1 \leq H_1$, $h_2 \leq H_2$, $k \leq P^{1/24}$: $kh_1h_2 \leq P^{1/24+1/48+1/24} = P^{5/48} \leq P^{1/8}$, so (C1)–(C3) hold with room $P^{-1/48}$. We prove

$$|T_2| \ll P^{23/24+\varepsilon} \quad \text{uniformly in } h_1, h_2, k;$$

then $|T_1| \ll P^{1-1/48+\varepsilon}$ and $|K_c| \ll P^{1-1/96+\varepsilon}$, the three savings balancing exactly at the chosen H_1, H_2 .

Step 2 (master identity; the trivial piece). By Lemma 5.1(iv), exactly,

$$\varphi_2 = \underbrace{(\Delta\Delta c)\theta_2}_{M_1} + \underbrace{(\Delta_2 c)(n+d_1)(\{W\} - \kappa_2)}_{M_2} + \underbrace{(\Delta_1 c)(n+d_2)(\{W'\} - \kappa'_2)}_{M_3} + \underbrace{c_{11}(\{\Delta\Delta Y\} - \kappa'' - \Delta_2\kappa_2)}_{M_4}.$$

By (E4), $|M_1| \leq 0.43kh_1h_2P^{-7/8}$, so $|e(M_1) - 1| \leq 2.7kh_1h_2P^{-7/8}$ and deleting M_1 changes T_2 by at most $2.7kh_1h_2P^{1/8} \leq 2.7P^{1/4}$ by (C1). No growing smooth part remains: every other bracket is bounded by 2.

Step 3 (the expansion inventory). The three remaining pieces are expanded as follows; all truncations, masses, and error costs are displayed here once and summed in Step 6.

(3a) M_2 , *sawtooth part* $(\Delta_2 c)(n+d_1)\{W\}$. The coefficient $B(n) = \Delta_2 c(n+d_1) \in (1.68, 1.85)kh_2P^{1/8}$ drifts by $|B'| \leq 0.22kh_2P^{-7/8}$ per step: freeze it on at most $2kh_2P^{1/4}+1$ windows on which it moves by $\leq P^{-1/8}$ (residual cost $\sum_n |e((B-B_0)\{W\}) - 1| \leq 6.3P^{-1/8} \cdot P = 6.3P^{7/8}$). Per window, Lemma 3.7 at the centre B_0 with $T = P^{1/2}/(2h_1)$, $J = P^{1/4}$ (hypothesis $T \geq 8(1+|B|)$: $P^{1/2}/(2h_1) \geq \frac{1}{2}P^{23/48}$ and $8(1+|B|) \leq 15kh_2P^{1/8} \leq 15P^{10/48}$, so the inequality holds for $P \geq P_0$): flat cost in total $\leq 8(1+1.85kh_2P^{1/8}) \cdot 2h_1P^{1/2} \leq 16h_1P^{1/2} + 30kh_1h_2P^{5/8} \leq 46P^{3/4}$ by (C1); the majorant Δ_J -part costs $\leq 2 \cdot 4P/J = 8P^{3/4}$ plus J -mode sums at coefficients $\leq 1/(J+1)$; modes $e(uW)$ with mass $\leq C \log P$ and $u \leq 1.85kh_2P^{1/8} + P^{1/2}/(2h_1)$, so that $uh_1 \leq 1.85P^{1/4} + P^{1/2}/2 \leq P^{1/2}$: every mode is a Lemma 5.2(i) object at shift $h_1 \leq P^{1/48}$. Window boundaries cost $\leq 2kh_2P^{1/4} \cdot 3.4P^{3/8} \leq 7P^{17/24}$.

(3b) M_2 , *carry part* $(\Delta_2 c)(n+d_1)\kappa_2$. Here $\kappa_2 \in \{0, 1\}$ is an integer, so exactly $e(-(\Delta_2 c)\kappa_2) = 1 + \kappa_2(e(-\Delta_2 c) - 1)$: the large coefficient never multiplies a sawtooth inside an exponential. The factor $e(-\Delta_2 c(n+d_1))$ is a smooth phase with $|(\Delta_2 c)''| \leq 2h_2 \sup |c'''| \leq 0.19kh_2P^{-15/8}$: class (D3). The weight $\kappa_2 = \{Y\} + \{W\} - \{Y+W\}$ (Lemma 5.1(ii)) is expanded *additively* as real numbers, each unit sawtooth by finite Fourier (Lemma 3.5) at truncation $J_2 = P^{1/24}$, coefficients $\leq 1/(\pi|q|)$, majorant cost $\leq 4P/J_2 = 4P^{23/24}$ per layer plus J_2 -mode averages. Since $Y+W = Y(n+d_1)$ exactly, the resulting modes are $e(qY(n))$, $e(qY(n+d_1))$ — Lemma 5.2(ii) objects — and $e(qW)$ — Lemma 5.2(i) objects.

(3c) M_3 . The mirror of (3a)–(3b) with $h_1 \leftrightarrow h_2$, producing $e(qY(n+d_2))$, $e(qW')$ -modes.

(3d) M_4 , *carries*. $\kappa'' = \{W\} + \{\Delta\Delta Y\} - \{W+\Delta\Delta Y\}$ with $W+\Delta\Delta Y = W(n+d_2)$ exactly, and $\Delta_2\kappa_2 = \kappa_2(n+d_2) - \kappa_2(n)$ with each κ_2 expanded as in (3b) at its base. The smooth factors $e(\mp c_{11})$ have $|c''| \leq$

$0.11kP^{-7/8}$: class (D3). New mode species: $e(q\{\Delta\Delta Y\}\text{-content})$ — slow modes, treated in Step 5 — and $e(qY(n+d))$ for $d \in \{0, d_2, d_1+d_2\}$, $e(qW(n+d_2))$: Lemma 5.2 objects again.

(3e) M_4 , *the anchor*. The factor $e(c_{11}\{\Delta\Delta Y\})$ is never expanded away. On the branch decomposition of Lemma 5.1(iii) (arcs absorbed by the exact shift device at $O(\log P)$ mass and $4P^{3/4}$ majorant, as in Theorem 4.4, Step 4), $\{\Delta\Delta Y\} = G - J_F$ with $G = F_\kappa(X-\theta)$ and $J_F = \lfloor G \rfloor$ frozen on the runs of (E6). The anchor phase $c_{11}(G - J_F)$ is present in **every** final piece; its curvature scale (E6) is what orders the whole classification of Step 5.

After (3a)–(3e), every final piece of T_2 has the exact form

$$e\left(\underbrace{c_{11}(G - J_F)}_{\text{anchor}} + \underbrace{\sum_i q_i Y(n+e_i)}_{\text{waves, } e_i \in \mathcal{D}} + \underbrace{uW\text{- and } u'W'\text{-modes}}_{\text{differenced waves}} + \underbrace{\text{slow modes} + \varphi}_{(\text{D3})\text{-smooth}}\right)$$

with at most one mode from each expansion layer (three layers), and piece weights whose total mass is $O(\log^3 P)$ beyond the displayed majorant and flat costs.

Step 4 (pieces with wave content: Lemma 5.2(ii)). Let $t = \sum_i q_i$ be the total wave frequency, $|t| \leq 3J_2 \leq 12P^{1/24}$, inside the coefficient budget of Lemma 5.2(ii).

If $t \neq 0$: telescoping, $\sum_i q_i Y(n+e_i) = tY(n+e_1) + \sum_{i \geq 2} q_i \Delta_{e_i-e_1} Y(n+\min(e_1, e_i))$, and after the targeted differencing of Lemma 5.2(ii) the differenced-wave remainders and the u, u' -modes become (D1) decorations, the anchor becomes the (D2) decoration, and the smooth content is (D3): Lemma 5.2(ii) applies and gives $\ll |t|^{-1/6} P^{23/24+\varepsilon}$ per piece. The weight sum is

$$\sum_{t \neq 0} \sum_{q_1+q_2+q_3=t} \frac{1}{\pi^3 \max(1, |q_1|) \max(1, |q_2|) \max(1, |q_3|)} |t|^{-1/6} \ll \sum_{t \geq 1} \frac{\log^2(2+t)}{t^{7/6}} \ll 1,$$

so the total wave-piece contribution is $\ll P^{23/24+\varepsilon}$.

If $t = 0$: the wave content collapses exactly to differenced waves — e.g. $q(Y(n+d_2) - Y(n)) = qW'(n)$ — i.e. to (D1)-type resonant decorations of the remaining piece; these are handled in Step 5. This exactness is the reason no near-resonant analysis is ever needed for the wave frequencies: the resonant combination is an identity, not an approximation.

Step 5 (pieces without wave content: the anchor classes). The remaining pieces carry the anchor, at most two differenced-wave modes $uW, u'W'$ (including the resonant remnants of Step 4, which by Lemma 5.1(iii) are smooth-per-branch with sub-unit θ -coefficients), slow modes, and (D3)-smooth content. Split by the anchor branch type.

(5a) Offset branches ($j \neq 0$). The anchor curvature is, by (E6) and the window-centre composite,

$$\lambda_a = \frac{729}{512} k|j| n^{-1/8} (1 + O(P^{-1/8})) \in [1.2, 1.5] k|j| P^{-1/8},$$

where $\frac{729}{512} = \frac{945}{512} - \frac{27}{64}$, by the following algebra. The smooth part of the anchor on an offset branch is $cF = \frac{3k}{4} \nu^{9/8} \cdot \frac{3}{2} j(m+\beta_1+\beta_2+\xi_1)^{1/2} = \frac{9}{8} k j \nu^{15/8} (1+O(P^{-1/4}))$ (Lemma 5.1(iii) and $m = \nu^{3/2}-\theta$), with second derivative $\frac{9}{8} \cdot \frac{15}{8} \cdot \frac{7}{8} k j \nu^{-1/8} = \frac{945}{512} k j \nu^{-1/8}$ up to the same relative error. The window-centre mode carries $u = -B(n_0)$, where $B = c \partial_m F = \frac{3k}{4} \nu^{9/8} \cdot \frac{3}{4} j m^{-1/2} = \frac{9}{16} k j \nu^{3/8} (1+O(P^{-1/4}))$ is the θ -sawtooth coefficient of the anchor (the θ -linear term of $cF(X-\theta)$), so its curvature contribution is $uX'' = -\frac{9}{16} \cdot \frac{3}{4} k j \nu^{-1/8} = -\frac{27}{64} k j \nu^{-1/8}$. The two terms have ratio $\frac{945}{512} : \frac{27}{64} = 4.375$, so the composite is single-signed at the displayed scale. Every competitor is dominated at a displayed margin: differenced-wave modes $\leq 0.84 u h_1 P^{-3/4} \leq 0.51 P^{-1/4}$ (since $u h_1 \leq 0.6 P^{1/2}$), ratio $\leq 0.43 P^{-1/8}$ against $\lambda_a \geq 1.2 P^{-1/8}$; resonant (D1) content $\leq 6 |q'| P^{-5/4}$ with $|q'| \leq 4 P^{1/24}$, ratio $\leq 20 P^{1/24-5/4+1/8} = 20 P^{-13/12}$; slow modes $\leq 3 J_2 |j| P^{-5/4}$, ratio $\leq 8 P^{1/24-9/8}$; (D3) content, ratio $\leq 3 h_1 h_2 P^{-1/2} \leq P^{-1/4}$. The θ -sawtooth of the anchor has coefficient $\frac{9}{16} k |j| P^{3/8}$ -scale with per-step drift $\leq 0.2 k |j| P^{-5/8} < 1$: at most $1.2 k |j| P^{3/8} + 1$ windows of length $\geq 0.8 P^{5/8} / (k |j|)$, window-boundary cost $\leq 1.2 k |j| P^{3/8} \cdot 0.92 (k |j|)^{-1/2} P^{1/16} \leq 1.1 (k |j|)^{1/2} P^{7/16}$. Off the collision band, window modes w are dominated at margin ≥ 4 either way and ride along or are estimated by Lemma 3.3 at their own scale

with $1/|w+B_0|$ weights ($\ll P^{7/8} \log P$ in total); on the collision band $|wX''| \in [\frac{1}{4}, 4]\lambda_a$, Lemma 3.8 with $(\alpha, \beta) = (\frac{15}{8}, \frac{3}{2})$ and $M \in [0.3, 6]k|j|P^{-1/8}$ gives $\leq C_E(|I_w|M^{1/2} + M^{-1/2} + (P/M)^{1/3})$ per window; summed over the at most $1.2k|j|P^{3/8} + 1$ windows with the $O(\log P)$ band mass,

$$\sum_w |I_w|M^{1/2} \leq 2.5 (k|j|)^{1/2} P^{15/16}, \quad \sum_w M^{-1/2} \leq 2.2 (k|j|)^{1/2} P^{7/16},$$

$$\sum_w (P/M)^{1/3} \leq 1.2k|j|P^{3/8} \cdot 1.5 (k|j|)^{-1/3} P^{3/8} = 1.8 (k|j|)^{2/3} P^{3/4} \leq 3.8 P^{1/36+3/4},$$

so the band total is $\ll (k|j|)^{1/2} P^{15/16} \log P$. The main estimate is Lemma 3.3 per frozen run at scale λ_a : run lengths $\geq \frac{1}{22} P^{1/4}/(|j|+1)$ (Lemma 5.1(iii)), and $\lambda_a^{-1/2} \leq 0.92(k|j|)^{-1/2} P^{1/16} \ll$ run length, so

$$\sum_{\text{runs}} (\ell \lambda_a^{1/2} + \lambda_a^{-1/2}) \leq 1.3 (k|j|)^{1/2} P^{15/16} + 21 (|j|+1) |j|^{-1/2} k^{-1/2} P^{13/16}.$$

Summed over the eight carry branches and $|j| \leq 3$ with the $O(\log^3 P)$ piece masses: $\ll (k)^{1/2} P^{15/16+\varepsilon} \leq 1.8 P^{1/48} P^{15/16+\varepsilon} = 1.8 P^{23/24+\varepsilon}$ at $k \leq P^{1/24}$ — the absorbed $(k|j|)^{1/2}$ loss, exactly at the bottleneck.

(5b) Zero-offset branches ($j = 0$). The expanded interpolant below gives the anchor curvature $\lambda_0 \in [1.0, 5.0] kh_1 h_2 P^{-5/8}$ (leading coefficient $\frac{243}{128}$, times the (E2) variation of the gaps and the dyadic range of $\nu^{-5/8}$). Runs of length $\geq \frac{1}{22} P^{3/4}/(h_1 h_2)$. Let $\mu = 0.84 \max(uh_1, u'h_2) P^{-3/4}$ be the strongest differenced-wave scale present. Three regimes.

- *Anchor-dominant* ($60\mu \leq \lambda_0$): Lemma 3.3 per run at λ_0 , all else dominated at margin ≥ 20 : $\leq 3.0(kh_1 h_2)^{1/2} P^{11/16} + 25 (h_1 h_2/k)^{1/2} P^{9/16}$.
- *Mode-dominant* ($\mu \geq 60\lambda_0$, i.e. $uh_1 \geq 60 kh_1 h_2 P^{1/8}$ -form): Lemma 5.2(i) with the undifferenced anchor as decoration: its run boundaries number $\leq 22h_1 h_2 P^{1/4} \leq 22P^{5/16}$, cost $\leq 22P^{5/16} \cdot 3.4(uh_1)^{-1/2} P^{3/8} \leq 75P^{11/16}$; its smooth part is dominated at margin ≥ 20 by hypothesis. On a zero-offset branch the anchor's θ -coefficient is $c \partial_m F = \frac{3}{4} c \cdot (-\frac{1}{2} \beta_1 \beta_2 (m+\xi)^{-3/2})$, hence $|B| \leq 1.2 kh_1 h_2 P^{-1/8} \leq 1.2 P^{-1/48} < 1$ by (C1): the sawtooth is sub-unit, Lemma 3.7 applies in a single window at $T = P^{1/2}$ (hypothesis $T \geq 8(1+|B|)$ is immediate), and there is no large- B window inventory. (The offset-scale amplitude $kh_1 h_2 P^{3/8}$ does not occur at $j = 0$.) Centre modes w are treated as in Lemma 5.2(i), Stage 5. Total $\ll ((uh_1)^{1/2} P^{5/8} + (h_1/u)^{1/2} P^{7/8} + P^{7/8}) P^\varepsilon$.
- *Middle band* ($\frac{1}{60} \leq \mu/\lambda_0 \leq 60$; coefficient mass $O(1)$ per layer): here up to three curvature scales meet, and this is the one place in the paper where the composite second derivative can cross zero *inside* cells. Since the cells are short ($\asymp P^{1/2}/h$) while the third-derivative scale is tiny, no inverse-power van der Corput term may be summed per cell; instead the transition set is measured once, against a global smooth model, and receives the trivial bound (Lemma 3.9).

Inventory. In this band $0.84 uh_1 P^{-3/4} = \mu \leq 60\lambda_0 \leq 300 kh_1 h_2 P^{-5/8}$ forces $u \leq 360 kh_2 P^{1/8} \leq 360 P^{5/24}$, and likewise $u' \leq 360 kh_1 P^{1/8} \leq 360 P^{5/24}$. The phase's second derivative f'' is smooth on the common refinement of the gap cells of both shifts (at most $1.5(h_1+h_2)P^{1/2} + 2 \leq 3.1P^{13/24}$), the anchor runs ($\leq 22h_1 h_2 P^{1/4} \leq 22P^{3/8}$), and the sawtooth windows already counted: in total at most $N \leq 3.5P^{13/24}$ pieces, for $P \geq P_0$. On each piece,

$$f'' = -\frac{9}{32} \left(uG_1(\nu+2h_1)^{-5/4} + u'G_2(\nu+2h_2)^{-5/4} \right) + (c_{11}(G_F - J_F))'' + wX'' + O(\rho_0\text{-small}),$$

with G_1, G_2, J_F (and the β_i inside G_F) frozen integers.

Interpolant. Write $\delta_h(\nu) = (\nu+2h)^{3/2} - \nu^{3/2}$ and $F_{\text{sm}}(\nu) = \frac{3}{4} (\Delta_1 X(\nu))(\Delta_2 X(\nu)) X(\nu)^{-1/2}$, the $j = 0$ branch function of Lemma 5.1(iii) with each frozen β_i replaced by the smooth gap $\Delta_i X$. The second derivative on a cell is

$$f'' = -\frac{9}{32} \left(uG_1(\nu+2h_1)^{-5/4} + u'G_2(\nu+2h_2)^{-5/4} \right) + (c_{11}(G_F - J_F))'' + wX'' + O(\rho_0\text{-small}),$$

with G_i, J_F frozen. Expand the anchor *before* interpolating: since J_F is constant on the run,

$$(c_{11}(G_F - J_F))'' = 2c'G'_F + cG''_F + c''(G_F - J_F).$$

The last summand is $O(|c''|)$ because $|G_F - J_F| < 1$. Define the interpolant by replacing the remaining frozen integers in this expanded expression:

$$\begin{aligned} \Lambda(\nu) = & -\frac{9}{32}u\delta_{h_1}(\nu)(\nu+2h_1)^{-5/4} - \frac{9}{32}u'\delta_{h_2}(\nu)(\nu+2h_2)^{-5/4} \\ & + 2c'(\nu)F'_{\text{sm}}(\nu) + c(\nu)F''_{\text{sm}}(\nu) + \frac{1}{2}c''(\nu) + wX''(\nu). \end{aligned}$$

The replacement $J_F \mapsto F_{\text{sm}} - \frac{1}{2}$ is made only in the already-expanded c'' -term, and produces the displayed $\frac{1}{2}c''$. It is *not* a replacement in the phase $c(G - J_F)$, which would collapse the anchor to $c/2$ and destroy the curvature.

Each replacement moves its argument by at most 1, so

$$|f'' - \Lambda| \leq \frac{9}{32}(u+u')P^{-5/4} + |c''_{11}| + 8k(h_1+h_2)P^{-9/8} \leq 219P^{-25/24} + 0.11P^{-5/6}.$$

The second term is the leading interpolant error and is not absorbed into a smaller multiple of $P^{-5/6}$. The first term dies relative to it. The comparison $V \geq 10|f'' - \Lambda|$ below is read from this two-term majorant, for $P \geq P_0$.

Leading monomials. Mean-value expansion gives $\delta_h(\nu) = 3h\xi^{1/2}$ with $\xi \in (\nu, \nu+2h)$, hence $\delta_h(\nu)(\nu+2h)^{-5/4} = 3h\nu^{-3/4}(1 + O(hP^{-1}))$, and

$$-\frac{9}{32}u\delta_{h_1}(\nu)(\nu+2h_1)^{-5/4} = -\frac{27}{32}uh_1\nu^{-3/4}(1 + O(h_1P^{-1})),$$

and likewise for u' . For the anchor, $\Delta_i X = 3h_i\nu^{1/2}(1 + O(h_iP^{-1}))$ and $X^{-1/2} = \nu^{-3/4}$, so $F_{\text{sm}} = \frac{27}{4}h_1h_2\nu^{1/4}(1 + O(hP^{-1}))$. With $c = \frac{3k}{4}\nu^{9/8}$,

$$2c'F'_{\text{sm}} + cF''_{\text{sm}} = \frac{243}{128}kh_1h_2\nu^{-5/8}(1 + O(hP^{-1})):$$

indeed $cF_{\text{sm}} = \frac{81k}{16}h_1h_2\nu^{11/8}(1 + O(hP^{-1}))$, so $(cF_{\text{sm}})'' = \frac{2673}{1024}kh_1h_2\nu^{-5/8}(1 + O(hP^{-1}))$ and $c''F_{\text{sm}} = \frac{729}{1024}kh_1h_2\nu^{-5/8}(1 + O(hP^{-1}))$, and the difference is $2c'F' + cF'' = (cF)'' - c''F$. The term $\frac{1}{2}c''$ is $O(kP^{-7/8})$ and is absorbed in the perturbation below. The window mode is exactly $wX'' = \frac{3}{4}w\nu^{-1/2}$.

Thus $\Lambda = \Phi'' + r$, where

$$\Phi(\nu) = a\nu^{5/4} + b\nu^{11/8} + w\nu^{3/2}, \quad a = -\frac{27}{10}(uh_1 + u'h_2), \quad b = \frac{81}{22}kh_1h_2,$$

and the remainder r collects the relative $O(hP^{-1})$ expansions, the $\frac{1}{2}c''$ term, and the $O(\rho_0)$ decorations. (Check: Φ'' has leading coefficients $a \cdot \frac{5}{4} \cdot \frac{1}{4} = -\frac{27}{32}(uh_1 + u'h_2)$ and $b \cdot \frac{11}{8} \cdot \frac{3}{8} = \frac{243}{128}kh_1h_2$.) The exponents $(\frac{5}{4}, \frac{11}{8}, \frac{3}{2})$ lie in E and are pairwise distinct. If $w = 0$, drop the third term and apply Lemma 3.8 (or Lemma 3.3 if only one of a, b is present). In all cases the perturbation satisfies the $\rho_0(E)$ bounds of Lemma 3.8/3.9 for $P \geq P_0$: each relative error is $O(P^{-1/4})$, and $|\frac{1}{2}c''|/B_{\text{lead}} = O(P^{-1/4})$ whenever $kh_1h_2 \geq 1$. The scale is

$$S = \max(|uh_1 + u'h_2|P^{-3/4}, kh_1h_2P^{-5/8}, |w|P^{-1/2}),$$

and the middle-band constraints give $1.0P^{-5/8} \leq S \leq 300P^{-1/2}$: the lower bound is the anchor when $kh_1h_2 \geq 1$, and the upper bound uses $kh_1h_2 \leq P^{1/8}$ from (C1) together with $\mu \leq 60\lambda_0 \leq 300kh_1h_2P^{-5/8}$ and the collision-band restriction $|wX''| \ll P^{-1/2}$.

Splitting. Choose $V := 3S^{1/2}P^{-11/24}$, so that $V/S \leq 6.7P^{-7/48}$ (hence $V \leq c_7S/2$ at $c_7 = 1/288$ for $P \geq P_0$) and $V \geq 1.35P^{-37/48} \geq 10|f'' - \Lambda|$ against the two-term interpolant majorant. The comparison $V \leq c_7S/2$ is the large ineffective threshold of the standing estimates: until it holds, a three-term zero of Φ'' can keep Ω_V of length $\Theta(P)$ on a dyadic block. The length bound below is

that of Lemma 3.9, not a claim that the sublevel is short at small P ; interval counts remain $O_E(1)$. By Lemma 3.9 the set $\Omega = \{\nu : |\Lambda(\nu)| \leq V\}$ is a union of at most $C(E)$ intervals of total length $\leq C(E) P(V/S)^{1/2} \leq 2.6 C(E) P^{89/96}$, and on its complement f'' is single-signed per interval with $0.9V \leq |f''| \leq 1.1 C(E) S$. The three costs:

$$\begin{aligned} \text{transition (trivial):} & \leq 2.6 C(E) P^{89/96}; \\ \text{piece boundaries:} & \leq (N+C(E)) (0.9V)^{-1/2} \leq 3.5 P^{13/24} \cdot 0.91 P^{37/96} + O_E(P^{37/96}) \leq 3.2 P^{89/96}; \\ \text{good pieces (Lemma 3.3):} & \sum \ell (1.1 C(E) S)^{1/2} \leq C'(E) P \cdot (300)^{1/2} P^{-1/4} \leq 18 C'(E) P^{3/4}. \end{aligned}$$

The middle band therefore totals $\leq C(E) P^{89/96} \log P \leq P^{15/16}$ for $P \geq P_0$.

In all three regimes the $j = 0$ pieces total $\ll P^{15/16+\varepsilon} \ll P^{23/24}$.

Step 6 (assembly). Additive costs: the M_1 deletion ($2.7P^{1/4}$); majorants (≤ 3 sawtooth layers at $4P/J_2 = 4P^{23/24}$ each, plus $8P^{3/4}$, $46P^{3/4}$, and the displayed window residuals $\leq 7P^{7/8}$); flat costs ($\leq 46P^{3/4}$ per layer). Multiplicative costs: mode masses over at most three expansion layers plus the shift devices, $O(\log^3 P) = P^\varepsilon$. Piece totals: Step 4, $\ll P^{23/24+\varepsilon}$; Step 5a, $\leq 1.8P^{23/24+\varepsilon}$; Step 5b, $\ll P^{15/16+\varepsilon}$; slow modes and (D3) remnants, $\ll P^{7/8+\varepsilon}$. Hence $|T_2| \ll P^{23/24+\varepsilon}$, and by Step 1

$$|T_1|^2 \leq 2P^{2-1/24} + CP^{1+23/24+\varepsilon} \Rightarrow |T_1| \ll P^{1-1/48+\varepsilon}, \quad |K_c|^2 \leq 2P^{2-1/48} + CP^{2-1/48+\varepsilon},$$

so $|K_c| \ll P^{1-1/96+\varepsilon}$. Exponents deliberately unoptimized. $\square$

Two remarks on the constants. First, the k -range is sharp for this assembly: the class-(5a) loss $(k|j|)^{1/2} P^{15/16}$ meets the wave-piece bottleneck $P^{23/24}$ exactly at $k = P^{1/24}$, which is why the theorem is uniform on the original range with no k -explicit statement needed. Second, the exponent $\frac{1}{96} = \frac{1}{4} \cdot \frac{1}{24}$ traces entirely to the depth-2 strength $P^{23/24}$ of the exact level-2 waves (Lemma 5.2(ii)); any improvement of the wave bound improves δ proportionally.

6. Application: depth four complete

Theorem 6.1 (the OOO^* splits; depth four complete over odd starts). For $w \in \{OOOE, OOOO\}$,

$$\#\{n \leq N \text{ odd} : \text{word}_4(n) = w\} = \frac{N}{16} + O(N^{1-1/96+\varepsilon}).$$

Hence every length-4 itinerary word class of *odd* starts — the eight words with first letter O — satisfies $\#\{n \leq N\} = 2^{-4}N + O(N^{1-\delta_w})$ with explicit $\delta_w > 0$: depth-4 parity equidistribution over odd starts is complete. (E-rooted words, i.e. even starts, are a different and easier problem — one square root drops the state to scale $\sqrt{N}$ — and are not treated in this paper.)

Proof. Step A (expansion; the mode ranges). The class indicator expands (Lemma 3.6, then Lemma 3.5 at truncation $J_3 = P^{1/96}$, majorant cost $4P/J_3 = 4P^{1-1/96}$ per layer) into mode sums

$$S_{ijk} = \sum_{\substack{n \sim P \\ n \text{ odd}}} e\left(\frac{i}{2}X + \frac{j}{2}Y + \frac{k}{2}v^{3/2}\right), \quad |i| \leq 2P^{1/96}, \quad |j| \leq 2P^{1/96}, \quad 1 \leq |k| \leq 2P^{1/96},$$

with weights $\leq \min(1, 2/|i|) \min(1, 2/|j|) \min(1, 2/|k|)$ of total mass $O(\log^3 P)$. The $k = 0$ sums are Theorems 4.1 and 4.4. Conjugating, take $k \geq 1$; then every displayed sign below is as written.

Step B (the depth-4 identity). Three applications of the Taylor pattern of Lemma 4.3(i), taken to second order, give the exact identity (odd $n \geq 5$)

$$v^{3/2} = -\frac{5}{64}n^{27/8} + \frac{9}{32}mn^{15/8} - \frac{45}{64}m^2n^{3/8} + \frac{15}{64}vn^{9/8} + \frac{45}{32}vmn^{-3/8} - \frac{9}{64}vm^2n^{-15/8} + \text{err},$$

$|\text{err}| \leq \frac{3}{4}n^{-9/8}$ (discarding err costs $\leq 2\pi k P^{-1/8} \cdot P \leq 7P^{7/8}$): the fourth-letter phase is a polynomial of degree $(2, 1)$ in the integer pair (m, v) with smooth coefficients. Substituting $m = X - \theta$ in the v -coefficients, the total smooth v -coefficient is *exactly*

$$c(\nu) := \frac{k}{2} \left(\frac{15}{64} + \frac{45}{32} - \frac{9}{64} \right) \nu^{9/8} = \frac{3k}{4} \nu^{9/8},$$

the monomial weight of Theorem 5.3, and the $v\theta$ - and $v\theta^2$ -cross coefficients are $\leq \frac{45}{64}k\nu^{-3/8} + \frac{9}{32}k\nu^{-3/8} \leq 1.2k\nu^{-3/8}$ and $\leq \frac{9}{128}k\nu^{-15/8}$: sub-unit and decaying. The pure- m part is $\mu_1 m + \mu_2 m^2$ with $\mu_1 = \frac{9}{64}k\nu^{15/8}$, $\mu_2 = -\frac{45}{128}k\nu^{3/8}$, and the pure-smooth part $-\frac{5}{128}k\nu^{27/8} + \mu_1 X + \mu_2 X^2$.

Step C (double differencing; corner exactness). Apply Step 1 of Theorem 5.3 ($H_1 = P^{1/48}$, $H_2 = P^{1/24}$; now $kh_1 h_2 \leq 2P^{1/96+1/48+1/24} \leq P^{1/8}$, so (C1)–(C3) hold) to the whole mode phase. On each level-1 carry branch of Lemma 5.1(iii), the four corner values obey the exact relations

$$m(n+d) = m + \beta_d, \quad \theta(n+d) = \theta + \delta_d(\nu) - \beta_d \quad (d \in \mathcal{D}),$$

with β_d frozen integers and δ_d smooth: the doubly differenced phase on a branch is an exact function $\Phi(\nu, \theta)$ of ν and the *single* sawtooth $\theta = \{X\}$. Two consequences are used repeatedly. First, the θ^2 -contents of the four corners cancel exactly in the $(+, -, -, +)$ pattern, so $|\partial_\theta^2 \Phi| \leq 3kh_1 h_2 P^{-13/8} + 4|\Delta\Delta\mu_2| \leq P^{-1}$: no quadratic sawtooth survives, and $e(\Phi) = e(\Phi(\nu, 0) + B(\nu)\theta)(1 + O(P^{-1}))$ with $B = \partial_\theta \Phi(\nu, 0)$. Second, the wave and kernel contents are exactly those of Theorem 5.3 with $c = \frac{3k}{4}\nu^{9/8}$: the v -linear part contributes $\Delta\Delta(c\theta_2)$ (the master identity, Steps 2–3 of Theorem 5.3) plus $\Delta\Delta(cY)$ -content whose wave frequencies join Step 4's bookkeeping.

Step D (passenger inventory). Every phase content beyond the bare kernel is listed here with its range and its check.

- *The $\frac{i}{2}X$ -passenger*, $|i| \leq 2P^{1/96}$: $\Delta\Delta(\frac{i}{2}X)$ is smooth with second derivative $\leq \frac{i}{2} \cdot 4h_1 h_2 \sup |X''''| \leq 2.3ih_1 h_2 P^{-5/2} \leq P^{-9/4}$: class (D3), dominated by every retained scale at margin $\leq P^{-3/2}$.
- *The $\frac{j}{2}Y$ -passenger*, $|j| \leq 2P^{1/96}$: $\Delta\Delta(\frac{j}{2}Y)$ is exactly four waves $\pm \frac{j}{2}Y(n+d)$, $d \in \mathcal{D}$, with total frequency 0. In Step 4 of Theorem 5.3 it shifts each wave coefficient q_d by at most $P^{1/96}$, keeping $|q_d| \leq 3P^{1/24} + P^{1/96} \leq 4P^{1/24}$ — inside Lemma 5.2(ii)'s coefficient budget — and leaves t unchanged; its $t = 0$ remnants are differenced waves that shift the u -coefficients of Step 5 by $\leq P^{1/96}$, harmless against the budget $uh_1 \leq 0.6P^{1/2}$.
- *The pure- m passengers* $\mu_1 m + \mu_2 m^2$: on a branch, by corner exactness, $\Delta\Delta(\mu m) = (\Delta\Delta\mu)m + (\Delta_2\mu)\beta_1 + (\Delta_1\mu)\beta_2 + \mu_{11}j$ and $\Delta\Delta(\mu m^2) = (\Delta\Delta\mu)m^2 + (\Delta_2\mu)(2\beta_1 m + \beta_1^2) + (\Delta_1\mu)(2\beta_2 m + \beta_2^2) + \mu_{11}(2jm + \beta_{12}^2 - \beta_1^2 - \beta_2^2)$, j the branch offset. The θ -coefficients that arise from $m = X - \theta$ in these terms are $|\Delta\Delta\mu_1| \leq 0.92kh_1 h_2 P^{-1/8} \leq P^{-1/48}$, $|2(\Delta_2\mu)\beta_1| \leq 2.3kh_1 h_2 P^{-1/8} \leq 3P^{-1/48}$, and $|2j\mu_2|$ -content, which joins $B(\nu)$ below; all sub-unit ones are expanded in the slow-mode families at $O(\log P)$ mass. The smooth parts join the two anchor families and are accounted in Step E.
- *The $v\theta$ -crosses*: writing $v = Y - \theta_2$ and linearizing $Y = m^{3/2}$ once more by Lemma 4.3(i), the crosses decompose into $\nu^{15/8}$ -scale θ -sawtooth content (joining $B(\nu)$), a $\theta\theta_2$ -product with coefficient $\leq 1.2k\nu^{-3/8}$ (deleted at cost $\leq 31kP^{5/8} \leq P^{2/3}$), and decaying remainders.

Step E (the two sign-critical composites, rerun). The passengers change the two composites of Step 5 of Theorem 5.3, and both are recomputed here from the smooth model of the *total* phase — the model in which every frozen integer is replaced by its interpolant, as in Step 5b there; each unit deviation of a frozen integer perturbs the curvatures below by $\leq Ck\nu^{-13/8}$, a relative $O(P^{-1})$.

Offset branches ($j \neq 0$). The model offset content is $\frac{k}{2}j\partial_m(m^{9/4}) = \frac{9k}{8}jm^{5/4}$, i.e. the smooth phase $\frac{9}{8}kj\nu^{15/8}$ with curvature $\frac{945}{512}kj\nu^{-1/8}$ — the same as in Theorem 5.3 — *but* the total θ -coefficient is now $B = \frac{k}{2}j\partial_m\partial_\theta$ -content $= -\frac{9k}{8}j \cdot \frac{5}{4}m^{1/4} = -\frac{45}{32}kj\nu^{3/8}(1 + O(P^{-1/4}))$, two and a half times the bare-kernel value $\frac{9}{16}kj\nu^{3/8}$. The window-centre mode therefore carries $u_0 X'' = -\frac{45}{32} \cdot \frac{3}{4}kj\nu^{-1/8} = -\frac{135}{128}kj\nu^{-1/8}$, and the composite is

$$\left(\frac{945}{512} - \frac{540}{512}\right)kj\nu^{-1/8} = \frac{405}{512}kj\nu^{-1/8}, \quad \text{ratio } 945 : 540 = 7 : 4,$$

single-signed with $\lambda'_a \in [0.72, 0.80]k|j|P^{-1/8}(1 + O(P^{-1/8}))$. The competitors of Step 5a, now read against $\lambda'_a \geq 0.72k|j|P^{-1/8} \geq 0.72P^{-1/8}$, remain dominated: differenced-wave modes $\leq 0.84uh_1 P^{-3/4} \leq 0.51P^{-1/4}$, ratio $\leq 0.71P^{-1/8}$; resonant (D1) content, ratio $\leq 7P^{-9/16}$; slow modes $\leq 3J_2|j|P^{-5/4}$, ratio $\leq 5P^{1/24-5/4+1/8} = 5P^{-13/12}$; (D3) content, ratio $\leq 4h_1 h_2 P^{-1/2} \leq P^{-1/4}$. The window count grows by $\frac{45/32}{9/16} = 2.5$, so at most $3.0k|j|P^{3/8} + 1$ windows; the collision-band Lemma 3.8 sum is then

$\leq 4.0 (k|j|)^{1/2} P^{15/16} \log P$. Run lengths are unchanged, and

$$\sum_{\text{runs}} (\ell \lambda_a'^{1/2} + \lambda_a'^{-1/2}) \leq 1.7 (k|j|)^{1/2} P^{15/16} + 26 (|j|+1) |j|^{-1/2} k^{-1/2} P^{13/16}.$$

Summed over the eight branches and $|j| \leq 3$ with the $O(\log^3 P)$ piece masses: $\ll k^{1/2} P^{15/16+\varepsilon} \leq P^{1/48} P^{15/16+\varepsilon} = P^{23/24+\varepsilon}$ already at the kernel's $k \leq P^{1/24}$; the actual range $k \leq 2P^{1/96}$ sits strictly inside.

Zero-offset branches ($j = 0$). The model $h_1 h_2$ -content of the total differenced phase is $4h_1 h_2 \cdot \frac{k}{2} \cdot \frac{27}{8} \cdot \frac{19}{8} \cdot \frac{11}{8} \cdot \frac{3}{8} \nu^{-5/8} = 8.27 k h_1 h_2 \nu^{-5/8} (1 + O(hP^{-1/2}))$, form; its curvature is

$$4h_1 h_2 \cdot \frac{k}{2} \cdot \frac{27}{8} \cdot \frac{19}{8} \cdot \frac{11}{8} \cdot \frac{3}{8} \nu^{-5/8} = 8.27 k h_1 h_2 \nu^{-5/8} (1 + O(hP^{-1/2})),$$

positive and single-signed, so $\lambda'_0 \in [4.5, 8.5] k h_1 h_2 P^{-5/8}$ replaces $\lambda_0 \in [0.2, 0.9] k h_1 h_2 P^{-5/8}$. The three-regime split of Step 5b is read at this scale. The thresholds (factor 60) are scale-free. In the anchor-dominant regime, Lemma 3.3 per run gives $\leq 4.0 (k h_1 h_2)^{1/2} P^{11/16} + 12 (h_1 h_2 / k)^{1/2} P^{9/16}$ (the first term grows like $\lambda_0'^{1/2}$, the second shrinks). In the mode-dominant regime the undifferenced-anchor decoration is the same as in Step 5b, and the $j = 0$ θ -coefficient remains sub-unit. In the middle band the interpolant of Step 5b is reused with B_{lead} replaced by $8.27 k h_1 h_2 \nu^{-5/8}$: the phase is still $\Phi = a\nu^{5/4} + b'\nu^{11/8} + w\nu^{3/2}$ with $b' = 8.27 \cdot \frac{64}{33} k h_1 h_2$, the $\rho_0(E)$ bounds are unchanged, and

$$S \leq \max(60\lambda'_0, \lambda'_0, |w|P^{-1/2}) \leq 520 P^{-1/2}$$

by (C1) and the collision-band restriction. The balanced choice $V = 3S^{1/2} P^{-11/24}$ still satisfies $V \leq c_7 S/2$ at $c_7 = 1/288$ and $V \geq 10|f'' - \Lambda|$ against the two-term interpolant majorant, for $P \geq P_0$. Transition length and piece-boundary costs shrink (larger S shrinks V/S and $V^{-1/2}$); good pieces cost $\leq P \cdot (520)^{1/2} P^{-1/4} \leq 23P^{3/4}$. The middle band therefore remains $\ll P^{15/16+\varepsilon}$.

The passengers of Step D are inside these estimates: the $\frac{i}{2}X$ -term is (D3) as listed; the $\frac{j}{2}Y$ -term shifts each q_d by at most $P^{1/96}$, keeping $|q_d| \leq 4P^{1/24}$ and $uh_1 \leq 0.6P^{1/2}$; the pure- m smooth pieces add a relative $O(P^{-1})$ to λ'_a and λ'_0 , already present in the $O(P^{-1/8})$ and $O(hP^{-1/2})$ factors; the sub-unit θ -coefficients of those pieces are among the slow modes already bounded.

Step F (assembly). With the inventory of Step D and the composites of Step E, Steps 2–6 of Theorem 5.3 give $|T_2| \ll P^{23/24+\varepsilon}$ uniformly in (i, j, k) on the stated ranges, hence $S_{ijk} \ll P^{1-1/96+\varepsilon}$. Summing the mode weights ($O(\log^3 P)$), the majorant costs ($\leq 4P^{1-1/96}$ per layer, three layers), and dyadic blocks gives the theorem. $\square$

The certified-descent density stays 13/16 at four steps — $OOO*$ is non-contracting at depth 4 ($3^3 > 2^4$). Deeper contracting words exist ($OOOEE$ and $OEOOE$ at length five are the next two), and the kernel theorem is plausibly the engine for counting them; but each such count requires a passenger inventory strictly larger than Step D's — including sawtooth coefficients at new scales and, at length seven, Taylor remainders that *grow* like $n^{9/32}$ and must be retained as subcritical phases rather than discarded — and we do not carry this out here. This paper claims no certified density beyond 13/16.

7. The Terras-style reduction and the frontier

The depth-by-depth counting assembles into a conditional Terras-style statement, and the reduction is unconditional:

Proposition 7.1 (equidistribution implies density-one descent). Let $d \geq 1$ and suppose that for every O -rooted itinerary word w of length d ,

$$|\#\{n \leq N : \text{word}_d(n) = w\} - 2^{-d}N| \leq E_d(N).$$

Then the starts with no contracting prefix of length $\leq d$ number at most

$$e^{-cd} N + 2^d E_d(N), \quad c = 2 \left(\frac{\log 2}{\log 3} - \frac{1}{2} \right)^2 > 0.0342.$$

Every other start $n \geq 2$ satisfies $J^t(n) < n$ for some $t \leq d$ with the uniform power-envelope certificate of Proposition 3.1. Consequently, if $E_d(N) = O_d(N^{1-\delta_d})$ with $\delta_d > 0$ for every d , the set of starts admitting a finite descent certificate has natural density 1.

Proof. Every E -rooted word has a contracting prefix at length one ($3^0 < 2$), so the starts with no contracting prefix of length $\leq d$ all realize an O -rooted word of length d . A word w of length d has a contracting prefix iff $3^{o_t} < 2^t$ for some $t \leq d$, where o_t counts odd letters among the first t . If w has no contracting prefix then $3^{o_d} \geq 2^d$, i.e. $o_d \geq \beta d$ with $\beta = \log 2 / \log 3 = 0.6309 \dots$. The number of such words is at most $2^d \Pr[\text{Bin}(d, \frac{1}{2}) \geq \beta d] \leq 2^d e^{-2(\beta-1/2)^2 d}$ by Hoeffding's inequality. Each such class has at most $2^{-d}N + E_d(N)$ members by the O -rooted hypothesis; summing gives the count. The density-one statement follows by letting $d \rightarrow \infty$ slowly with N (any $d(N) \rightarrow \infty$ with $2^d E_d(N)/N \rightarrow 0$). $\square$

Sections 4–6 prove the hypothesis at every depth $d \leq 4$, so the conclusion of Proposition 7.1 is unconditional for those depths. The first open case is depth 5: the $OOOO*$ split. It has an exact shape, one nesting deeper than Theorem 5.3. Write $Z = v^{3/2}$, $z = \lfloor Z \rfloor$, $\theta_3 = Z - z$; after four odd letters the fifth is the parity of $\lfloor z^{3/2} \rfloor$.

Lemma 7.2 (level-3 kernel reformulation). For odd $n \geq 5$,

$$\frac{1}{2}(v^{9/4} - z^{3/2}) - \frac{3}{4}z^{1/2}\theta_3 = R_3, \quad 0 \leq R_3 \leq \frac{3}{16}z^{-1/2}.$$

Consequently the fifth-letter phase is, up to $kR_3 \ll kn^{-27/16}$ per term, the exponential sum of the level-3 local floor defect, with coefficient $c = \frac{3k}{4}z^{1/2} \asymp kn^{27/16}$: Lemma 5.1(i) with (m, v) replaced by (v, z) .

Proof. Taylor of $(z + \theta_3)^{3/2}$ at z , with $Z^{3/2} = v^{9/4}$. $\square$

For smooth c with $c \asymp kP^{27/16}$ and $c' \asymp kP^{11/16}$ on $n \sim P$ (the $z^{1/2}$ -shaped family), define

$$K_3(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e(c(n) \{ \lfloor \lfloor n^{3/2} \rfloor^{3/2} \rfloor^{3/2} \}).$$

Conjecture 7.3 (level-3 kernel cancellation). $K_3(P) \ll P^{1-\delta}$ for some $\delta > 0$, uniformly over the family above with $k \leq P^\varepsilon$.

The boundary between Theorem 5.3 and Conjecture 7.3 is quantitative and sharper than a derivative count. The smooth model iterates: where Theorem 5.3 used two Weyl differencings against $Y'' \asymp P^{1/4} \gg 1 > P^{-3/4} \asymp Y'''$, the level-3 model $n^{27/8}$ has $G''' \asymp P^{3/8} \gg 1 > P^{-5/8} \asymp G^{(4)}$ and predicts three. But the prediction does not descend to the nested floors. The kernel weight now has $c' \asymp kP^{11/16} \gg 1$, so no drift-1 interval exists for any expansion of the weight itself; the branch decomposition of Lemma 5.1(iii) has no analogue at the v -level, because v jumps by $\asymp n^{5/4}$ per step and the branch set is a *product* of two carry lattices, not a copy of one; and the forced inner linearization $v^{3/2} = m^{9/4} - \frac{3}{2}m^{3/4}\theta_2 + E_2$ trades θ_3 for a sawtooth family at coefficient scale $kn^{45/16}$, *above* the $9/4$ threshold where every method of this paper stops.

What survives at the frontier is a model problem and one theorem about it. Stripping every problem-specific structure (carries, defects, nesting) from K_3 leaves the *amplitude-product* sums

$$S = \sum_{t \leq L} e(A(t) \{B(t)\}), \quad 1 \ll A' \ll A,$$

with smooth monomial-type A, B (the instance above has $A \asymp P^{27/16}$, $A' \asymp P^{11/16}$). For $A' \ll 1$ partial summation makes the amplitude a tame passenger and the classical single-floor machinery applies; for $A' \gg 1$ we know of no nontrivial deterministic bound by any method. What can be proved is an L^2 identity in the shift, a one-page computation that uses no harmonic analysis:

Proposition 7.4 (shift-averaged L^2 bound). Let $A_1 < \dots < A_L$ be reals with $|A_t - A_{t'}| \geq A'_{\min}|t - t'|$ for some $A'_{\min} \geq 1$, and let $x_1, \dots, x_L$ be arbitrary reals. For $S_\lambda = \sum_{t \leq L} e(A_t \{x_t + \lambda\})$,

$$\left| \int_0^1 |S_\lambda|^2 d\lambda - L \right| \leq \frac{6}{\pi} \frac{L}{A'_{\min}} (\log L + 1).$$

Consequently, for any $\varepsilon \in (0, 1)$, Markov's inequality gives

$$|S_\lambda| \leq \sqrt{\frac{L}{\varepsilon} \left(1 + \frac{6}{\pi} \frac{\log L + 1}{A'_{\min}}\right)}$$

outside a shift set of measure at most ε . This is square-root cancellation times $\sqrt{\log L}$ in general, and genuine square-root cancellation when $A'_{\min} \gg \log L$ — which holds in the instance above, where $A'_{\min} \asymp P^{11/16}$.

Proof. Expand the square; the diagonal gives L . For $t \neq t'$, the function $\varphi(\lambda) = A_t\{x_t + \lambda\} - A_{t'}\{x_{t'} + \lambda\}$ is piecewise linear on $[0, 1)$ with *real* slope $A_t - A_{t'}$ on at most three arcs (jumps at $1 - \{x_t\}$ and $1 - \{x_{t'}\}$); on each arc $|\int e(\varphi) d\lambda| \leq 1/(\pi|A_t - A_{t'}|)$. Summing, $\sum_{t \neq t'} 3/(\pi A'_{\min}|t - t'|) \leq (6/\pi)(L/A'_{\min})(\log L + 1)$. For the second display, the measure of $\{|S_\lambda|^2 > T\}$ is at most $T^{-1}(L + (6/\pi)(L/A'_{\min})(\log L + 1))$; take $T = (L/\varepsilon)(1 + (6/\pi)(\log L + 1)/A'_{\min})$. $\square$

Two cautions. The amplitude separation $A' \gg 1$ — the very property that defeats every character expansion, since a Fourier window centred at the amplitude drifts by A' harmonics per step — is what makes the shift *average* trivial; but an average over shifts says nothing about the single shift $\lambda = 0$, which is the deterministic sum. Proposition 7.4 does not make the level-3 kernel “generically cancelling” in any sense that bears on Conjecture 7.3; it only locates the conjecture's difficulty as a specific-point problem inside a metric statement.

Conjecture 7.5 (the pure amplitude-product model). $|S| \leq L^{1-\delta}$ for some $\delta > 0$, for smooth monomial-type A, B with $1 \ll A' \ll A$ as above.

Three structural facts locate the difficulty of the de-randomization. No second averaging variable exists: amplitude separation forces any two sample points with $|A(p) - A(q)| \leq 1$ to coincide, and every family average available in the application re-enters either the differenced-kernel class or the amplitude-product class itself. The inverse theory is self-similar: any concentration or discrepancy inverse for $A\{B\} \bmod 1$ is a statement about $\sum e(jA\{B\})$ — the same class with amplitude jA — so the class is closed under its own inverse theorems and no bootstrapping is possible. And the metric statement does not transfer: $|dS_\lambda/d\lambda| \leq 2\pi A_{\max} L$ almost everywhere, so S_λ decorrelates at shift scale $1/A_{\max}$, and an almost-all- λ theorem leaves $\asymp \varepsilon A_{\max}$ exceptional cells among $\asymp A_{\max}$ — no measure argument pins $\lambda = 0$. The deterministic content of Conjecture 7.5 is a specific-point-in-metric-theory problem, the same species as the normality of $\sqrt{2}$: a class whose known successes all use special arithmetic.

Two analytic shortcuts around Conjecture 7.3 fail at recorded points, and we state them once: composing gap cells across two levels fails because on a cell where the first-level gap is constant the second-level gap still takes a new value at essentially every point, so no usable sub-cell survives; and reindexing by the image m strips one nesting level but introduces a fiber indicator of density $\asymp m^{-1/3}$, whose sawtooth requires savings beyond the exponent $1/3$ while the engine saves only $1/24$.

The open question, stated once:

It is open whether almost every odd-to-odd start has a finite descent certificate. By Proposition 7.1 that would follow from all-depth parity equidistribution, which is now a theorem through depth four; the first open case is the *OOOO** kernel of Conjecture 7.3, whose deterministic model instance is Conjecture 7.5.

8. Software note

A repository accompanies the paper: https://github.com/sneakyweasel/balanced_ternary/. It is not required to read or check any proof above. It contains Lean formalizations of the exact floor identities cited in Section 1.1 and of the companion manuscript's finite-word theorems, scaled-integer validations of the linearization identities of Lemmas 4.3, 4.6, 5.1, and 7.2, and exact-phase numerical probes of the sums K_c , K_3 , and the differenced sums of Lemma 5.2 and Theorem 5.3, all of which exhibit cancellation at square-root scale — stronger than the theorems claim and stronger than they need. The same companion records scaled-integer gates for the exact identities of Section 1.1 and the named numerical margins of Section 5 (Lemma 3.7 / Lemma 5.2 window hypotheses, the Step 5b interpolant majorant, and the Lemma 5.2 coefficient budget).

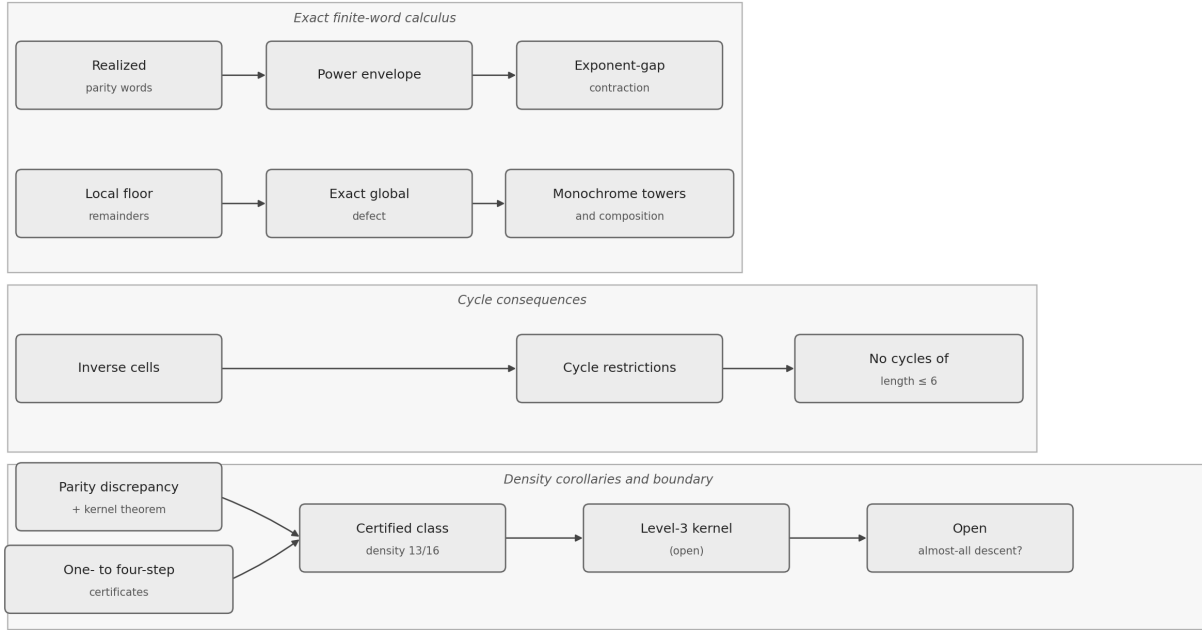


Figure 1: The theorem flow of the paper. The exact finite-word calculus of the companion manuscript feeds the contraction certificates; the discrepancy calculus with the kernel theorem counts every O-rooted itinerary class through depth four (certified-descent density $13/16$), leaving the level-3 kernel — and with it almost-all descent — open.

Probes and validations are checks, not proofs, and no statement in this paper depends on them. In particular they are not evidence for Conjecture 7.3 or Conjecture 7.5.

Acknowledgments

I used large language models extensively while drafting and revising the text, organizing companion notes, and as an interactive assistant for Lean statements, tests, and literature records. The models are not authors. The theorems of this paper are human proofs from the cited classical inequalities; the Lean certificates cover only the exact floor identities and the companion’s finite-word theorems. I take full responsibility for the contents.

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